

Downregulation of Semaphorin 4A in keratinocytes reflects the features of non-lesional psoriasis

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This paper advances an **important** new concept in psoriasis pathogenesis and implicates Sema4a as a homeostatic regulator that is highly epithelial-specific. The findings are **convincing** and lend support for the biology described here as a mechanism with therapeutic implications.

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Abstract

Psoriasis is a multifactorial disorder mediated by IL-17-producing T cells, involving immune cells and skin-constituting cells. Semaphorin 4A (Sema4A), an immune semaphorin, is known to take part in T helper type 1/17 differentiation and activation. However, Sema4A is also crucial for maintaining peripheral tissue homeostasis and its involvement in skin remains unknown. Here, we revealed that while Sema4A expression was pronounced in psoriatic blood lymphocytes and monocytes, it was downregulated in the keratinocytes of both psoriatic lesions and non-lesions compared to controls. Imiquimod application induced more severe dermatitis in Sema4A knockout (KO) mice compared to wild-type (WT) mice. The naïve skin of Sema4AKO mice showed increased T cell infiltration and IL-17A expression along with thicker epidermis and distinct cytokeratin expression compared to WT mice, which are hallmarks of psoriatic non-lesions. Analysis of bone marrow chimeric mice suggested that Sema4A expression in keratinocytes plays a regulatory role in imiquimod-induced dermatitis. The epidermis of psoriatic non-lesion and Sema4AKO mice demonstrated mTOR complex 1 upregulation, and the application of mTOR inhibitors reversed the skewed expression of

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cytokeratins in *Sema4AKO* mice. Conclusively, *Sema4A*- mediated signaling cascades can be triggers for psoriasis and targets in the treatment and prevention of psoriasis.

Introduction

While the infiltration of immune cells into skin plays a critical role in the development of psoriasis, as evidenced by interleukin (IL) -23/IL-17 axis (Fitch, Harper, Skorcheva, Kurtz, & Blauvelt, 2007; Hawkes, Yan, Chan, & Krueger, 2018; Kim & Krueger, 2017), recent studies have revealed that cells constructing skin structure, such as keratinocytes, fibroblasts, and endothelial cells, also play pivotal roles in the development (Heidenreich, Röcken, & Ghoreschi, 2009; Lowes, Russell, Martin, Towne, & Krueger, 2013; Zhang et al., 2023) and maintenance (Arasa et al., 2019; Francis et al., 2024; Q. Li et al., 2023; Ma et al., 2023; Tan et al., 2015; Zhu et al., 2020) of psoriasis. Among these cells, keratinocytes function as a barrier and produce cytokines, chemokines, and antimicrobial peptides against foreign stimuli, resulting in the activation of immune cells (Ni & Lai, 2020; Zhou, Chen, Cui, Shi, & Guo, 2022).

Semaphorins were initially identified as guidance cues in neural development (Kolodkin, Matthes, & Goodman, 1993; Pasterkamp & Kolodkin, 2003) but are now regarded to play crucial roles in other physiological processes including angiogenesis (Iragavarapu-Charyulu, Wojcikiewicz, & Urdaneta, 2020; Serini, Maione, Giraudo, & Bussolino, 2009), tumor microenvironment (Hui, Tam, Jiao, & Ong, 2019; Jiang et al., 2022; Nakayama, Kusumoto, Nakahara, Fujiwara, & Higashiyama, 2018; Rajabinejad et al., 2020), and immune systems (Garcia, 2019; Kanth, Gairhe, & Torabi-Parizi, 2021; M. Naito & Kumanogoh, 2023). Semaphorin 4A (*Sema4A*), one of the immune semaphorins, plays both pathogenic and therapeutic roles in autoimmune diseases (Cavalcanti et al., 2020; He et al., 2023), allergic diseases (Maeda et al., 2019), and cancer (Iyer & Chapoval, 2018; Liu et al., 2018; Y. Naito et al., 2023; Pan, Wang, & Ye, 2016). While *Sema4A* expression on T cells is essential for T helper type 1 differentiation in the murine *Propionibacterium acnes*-induced inflammation model and delayed-type hypersensitivity model (Kumanogoh et al., 2005), *Sema4A* amplifies only T helper type 17 (Th17)-mediated inflammation in the effector phase of murine experimental autoimmune encephalomyelitis (Koda et al., 2020). Accordingly, in multiple sclerosis, high serum *Sema4A* levels correlate with the elevated serum IL-17A, earlier disease onset, and increased disease severity (Koda et al., 2020; Nakatsuji et al., 2012). In anti-tumor immunity, research involving human samples and murine models suggests that *Sema4A* expressed in cancer cells and regulatory T cells promotes tumor progression (Delgoffe et al., 2013; Liu et al., 2018; Pan et al., 2016), while other reports reveal that *Sema4A* in cancer cells and dendritic cells bolsters anti-tumor immunity by enhancing CD8 T cell activity (Y. Naito et al., 2023; Suga et al., 2021). In addition to these roles in immune reactions, mice with a point mutation in *Sema4A* develop retinal degeneration, suggesting that *Sema4A* is also crucial for peripheral tissue homeostasis (Nojima et al., 2013). These reports suggest that the role of *Sema4A* can differ based on the disease, phase, and involved cells.

Herein, we investigated the roles of *Sema4A* in the pathogenesis of psoriasis by analyzing skin and blood specimens from psoriatic subjects and using a murine model.

Results

Epidermal *Sema4A* expression is downregulated in psoriasis

The analysis of previously-published single-cell RNA-sequencing data from control (Ctl) and psoriatic lesional (L) skin specimens (Kim et al., 2023) revealed detectable expression of *SEMA4A* in keratinocytes, dendritic cells, and macrophages in both Ctl and L. *SEMA4A* expression

was low in neural crest-like cells, fibroblasts, CD4 T cells, CD8 T cells, NK cells, and plasma cells, making the comparison of expression levels impractical (**Figure 1A and B**; **Figure 1**-figure supplement 1A-C). Dendritic cells and macrophages showed comparable *SEMA4A* expression levels between Ctl and L (**Figure 1C**). The adjusted *p*-value (*padj*) for *SEMA4A* in keratinocytes between Ctl and L was 2.83×10^{-39} , indicating a statistically significant difference despite not being visually prominent in the volcano plot, which shows comprehensive differential gene expression in keratinocytes (**Figure 1C**; **Figure 1**-figure supplement 1D).

Immunohistochemistry of Ctl and psoriasis demonstrated Sema4A expression in keratinocytes (**Figure 1D**). The staining intensity of Sema4A in epidermis was significantly lower in both non-lesions (NL) and L than in Ctl (**Figure 1E**). Relative mRNA expression of *SEMA4A* was also decreased in the epidermis of NL and L compared to Ctl, while it remained comparable in the dermis (**Figure 1F**). In contrast, the proportions of Sema4A-positive cells were significantly higher in blood CD4 and CD8 T cells, and monocytes in psoriasis compared to Ctl (**Figure 1G**; **Figure 1**-figure supplement 2). Serum Sema4A levels, measured by enzyme-linked immunosorbent assay (ELISA), were comparable between Ctl and psoriasis (**Figure 1H**). These findings demonstrate that the expression profile of Sema4A in psoriasis varies across cell types.

Psoriasis-like dermatitis is augmented in Sema4AKO mice

When psoriasis-like dermatitis was induced in wild-type (WT) mice and Sema4A knockout (KO) mice by imiquimod application on ears (**Figure 2A**), ear swelling on day 4 was more pronounced in Sema4AKO mice (**Figure 2B**) with upregulated *Il17a* gene expression (**Figure 2C**). Flow cytometry analysis of cells isolated from the ears revealed increased proportions of $V\gamma 2^+$ T cells, $V\gamma 2^+V\gamma 3^-$ double-negative (DN) $\gamma\delta$ T cells, and IL-17A-producing cells of those fractions in Sema4AKO epidermis (**Figure 2D and E**; **Figure 2**-figure supplement 1). In Sema4AKO dermis, there was also an increase in the proportions of $V\gamma 2^+$ T cells and IL-17A-producing $V\gamma 2^+$ T cells (**Figure 2D and E**). These results suggest that Sema4A deficiency in mice accelerates psoriatic profile. IL-17A-producing T cells in skin-draining lymph nodes (dLN) remained comparable between WT mice and Sema4AKO mice (**Figure 2F**).

Though the imiquimod model is well-established and valuable murine psoriatic model (van der Fits et al., 2009), the vehicle of imiquimod cream can activate skin inflammation that is independent of toll-like receptor 7, such as inflammasome activation, keratinocyte death and interleukin-1 production (Walter et al., 2013). This suggests that the imiquimod model involves complex pathway. Therefore, we subsequently induced IL-23-mediated psoriasis-like dermatitis (Figure2-figure supplement 2A), a much simpler murine psoriatic model, because IL-23 is thought to play a central role in psoriasis pathogenesis (Krueger et al., 2007; Lee et al., 2004). Although ear swelling on day 4 was comparable between WT mice and Sema4AKO mice (Figure2-figure supplement 2B), the epidermis, but not the dermis, was significantly thicker in Sema4AKO mice compared to WT mice (Figure2-figure supplement 2C). We found that the proportion of CD4 T cells among T cells was significantly higher in Sema4A KO mice compared to WT mice, while the proportion of $V\gamma 2$ and $DN\gamma\delta$ T cells among T cells was comparable between them (**Figure 2**-figure supplement 2D). On the other hand, focusing on IL-17A-producing cells, the proportion of IL-17A-producing $V\gamma 2$ and $DN\gamma\delta$ T cells in CD3 fraction in the epidermis was significantly higher in Sema4A KO mice, consistent with the results from imiquimod-induced psoriasis-like dermatitis. (**Figure 2**-figure supplement 2E).

Sema4A in keratinocytes may play a role in preventing murine psoriasis-like dermatitis

To investigate the cells responsible for the augmented ear swelling in Sema4AKO mice, bone marrow chimeric mice were next analyzed (**Figure 3A**). Since it has already been reported that bone marrow cells contain keratinocyte stem cells (Harris et al., 2004; Wu, Zhao, & Tredget, 2010), we confirmed that epidermis of mice deficient in non-hematopoietic Sema4A (WT $\rightarrow$ KO)

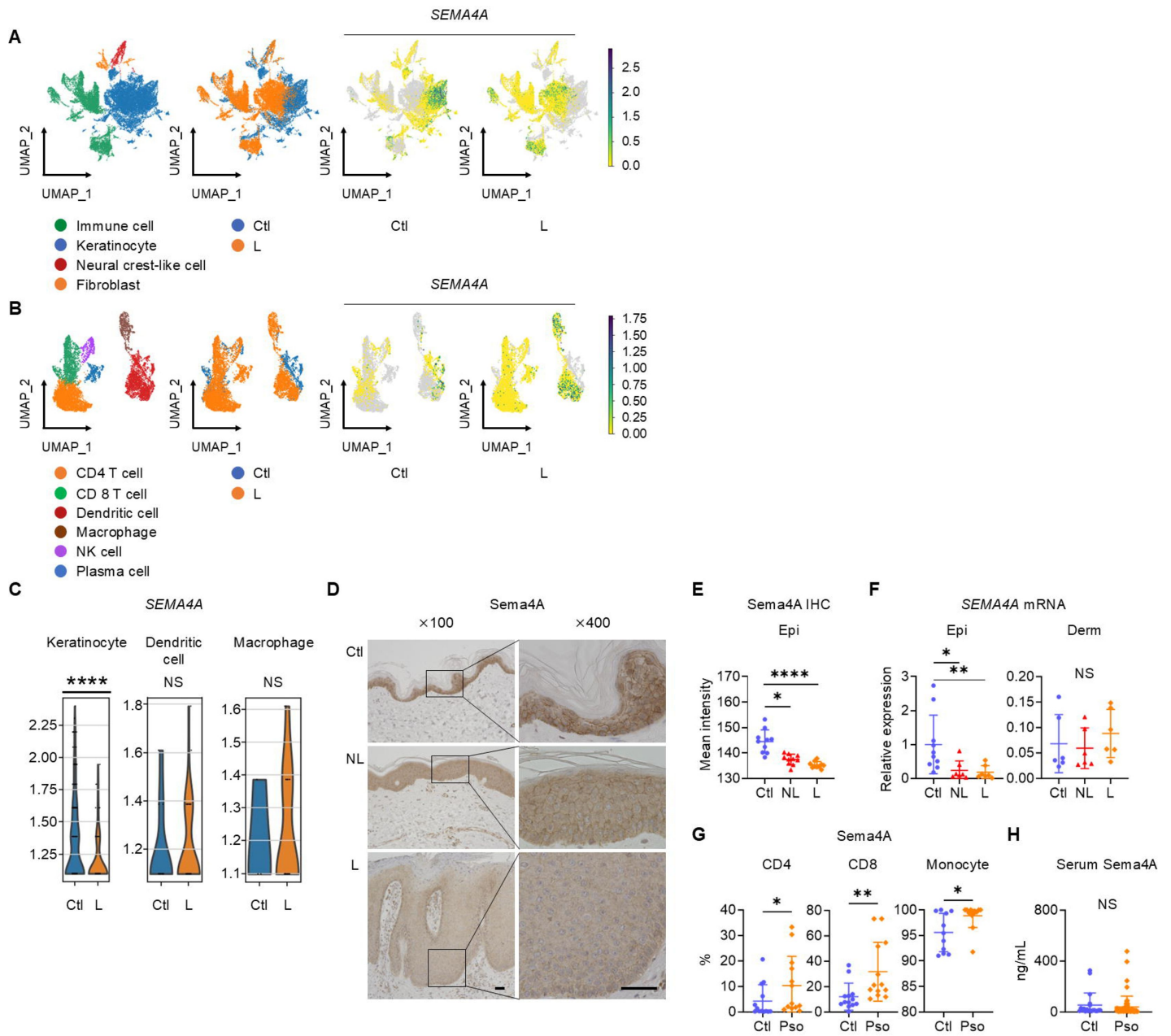


Figure 1

Epidermal Sema4A expression is downregulated in psoriasis.

(A) UMAP plots, generated from single-cell RNA-sequencing data (GSE220116), illustrate cell distributions from control (Ctl) and psoriatic lesion (L) samples ($n = 10$ for Ctl, $n = 11$ for L). (B) Subclustering of immune cells. (C) *SEMA4A* expression in keratinocytes, dendritic cells, and macrophages. **** $p < 0.0001$. NS, not significant. Analyzed using Python and cellxgene VIP. (D) Representative immunohistochemistry and magnified views showing Sema4A expression in Ctl, psoriatic non-lesion (NL), and L. Scale bar = 50 μ m. (E) Mean epidermal (Epi) Sema4A intensity in immunohistochemistry ($n = 10$ per group). Each dot represents the average intensity from 5 unit areas per sample. (F) Relative *SEMA4A* expression in Epi ($n = 10$ for Ctl, L, $n = 7$ for L and NL) and dermis (Derm, $n = 6$ per group). (G) Proportions of Sema4A-expressing cells in blood CD4 T cells (left), CD8 T cells (middle), and monocytes (right) from Ctl and psoriatic (Pso) patients. ($n = 13$ per group in CD4 and CD8, $n = 11$ for Ctl and $n = 13$ for Pso in Monocytes). (H) Serum Sema4A levels in Ctl ($n = 20$) and Pso ($n = 60$). E to H: * $p < 0.05$, ** $p < 0.01$, **** $p < 0.0001$. NS, not significant. **Figure 1** [source data 1](#) Excel file containing quantitative data for **Figure 1**.

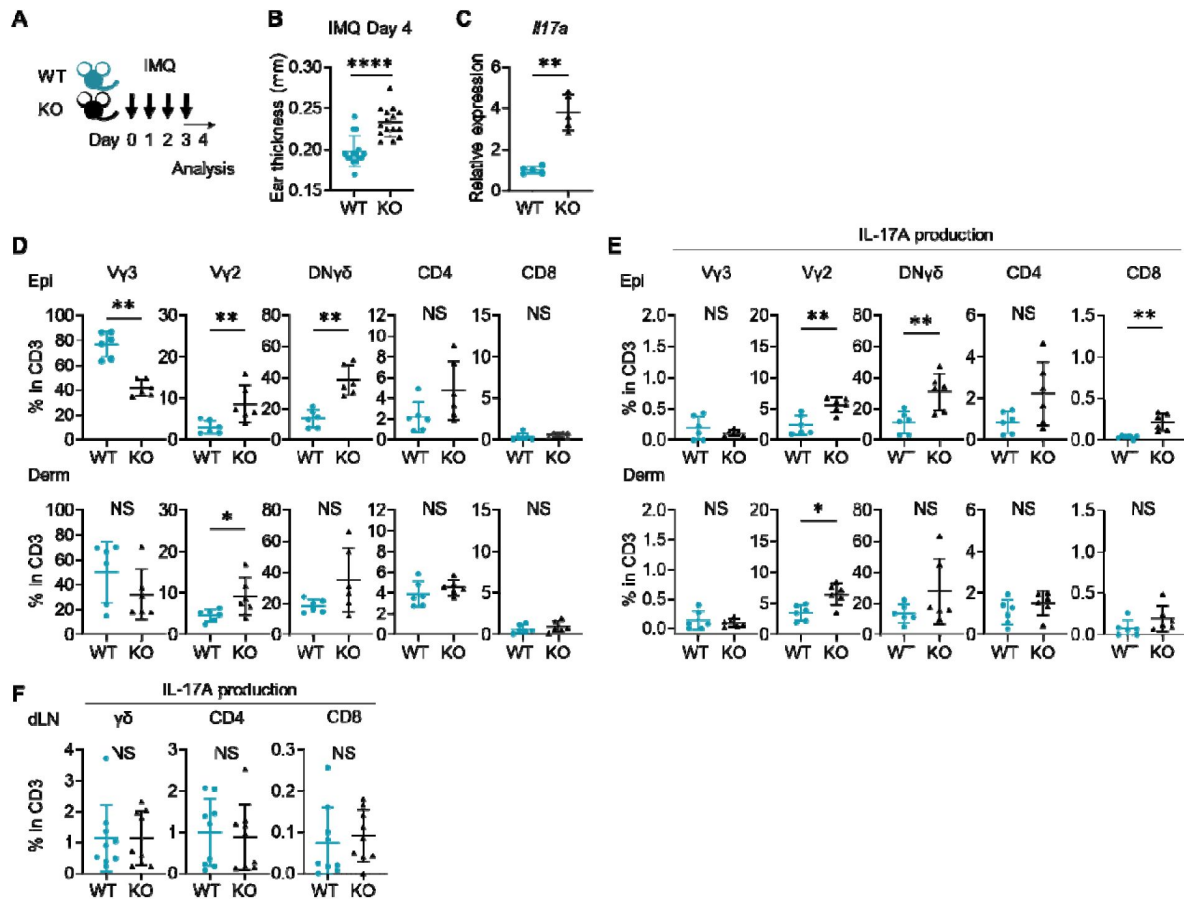


Figure 2

Imiquimod-induced Psoriasis-like dermatitis is augmented in Sema4AKO mice.

(A) Experimental scheme. Wild-type (WT, green) mice and Sema4A knockout (KO, black) mice were treated with 10 mg/ear of 5% Imiquimod (IMQ) for 4 consecutive days. Samples for flow cytometry analysis were collected on Day 4. (B) Ear thickness of WT mice and KO mice on Day 4 ($n = 15$ per group). (C) Relative expression of *Il17a* in epidermis ($n = 5$ per group). (D and E) The percentages of Vγ3, Vγ2, Vγ2-Vγ3-γδ (DNγδ), CD4, and CD8 T cells (D) and those with IL-17A production (E) in CD3 fraction in the Epi (top) and Derm (bottom) of WT and KO ears ($n = 6$ per group, each dot represents the average of 4 ear specimens). (F) The percentages of IL-17A- producing γδ, CD4, and CD8 T cells in CD3 fraction in skin-draining lymph nodes (dLN) ($n = 9$ per group). B-F: * $p < 0.05$, ** $p < 0.01$, **** $p < 0.0001$. NS, not significant. **Figure 2** -source data 1 Excel file containing quantitative data for **Figure 2**.

showed no obvious detection of *Sema4a*, thereby ruling out the impact of donor-derived keratinocyte stem cells infiltrating the host epidermis (**Figure 3** [figure supplement 1A](#)). WT → KO mice displayed more pronounced ear swelling than mice with intact *Sema4A* expression (WT → WT) following imiquimod application (**Figure 3B** [figure supplement 1B](#)). Similarly, mice with a systemic deficiency of *Sema4A* (KO → KO) showed severe ear swelling compared to mice deficient in hematopoietic *Sema4A* (KO → WT) (**Figure 3B** [figure supplement 1B](#)). Ear swelling was comparable between WT → WT mice and KO → WT mice (**Figure 3B** [figure supplement 1B](#)). Flow cytometry analysis revealed increased infiltration of IL-17A-producing DN γ 8 T cells in the epidermis, as well as V γ 2⁺ T cells and IL-17A-producing V γ 2⁺ T cells in the dermis, in WT → KO mice compared to WT → WT mice (**Figure 3C** [figure supplement 1B](#)). These findings suggest that non-hematopoietic cells, possibly keratinocytes, are primarily responsible for the increased imiquimod-induced *Sema4A* KO mice ear swelling.

Sema4AKO epidermis is thicker than WT epidermis with increased γ 8 T17 infiltration

Even without imiquimod application, *Sema4AKO* ears turned out to be slightly but significantly thicker than WT ears on week 8 while their appearance remained normal (**Figure 4A** [figure supplement 1A](#)). While epidermal thickness of back skin was comparable at birth (**Figure 4B** [figure supplement 1A](#)), on week 8, epidermis of *Sema4AKO* back and ear skin was notably thicker than that of WT mice (**Figure 4B** [figure supplement 1A](#)), suggesting that acanthosis in *Sema4AKO* mice is accentuated post-birth. Dermal thickness remained comparable between WT mice and *Sema4AKO* mice at both times (**Figure 4B** [figure supplement 1A](#)). The epidermis of WT ear at week 8 showed significantly higher *Sema4a* mRNA expression compared to the dermis (**Figure 4C** [figure supplement 1A](#)). Based on these observations, *Sema4A* appears to play a more pronounced role in epidermis than in dermis.

Sema4AKO epidermis exhibited increased expression of *Ccl20*, *Tnfa*, and *Il17a* and a trend of upregulation of *S100a8* compared to WT epidermis (**Figure 4** [figure supplement 1A](#)). These differences were not observed in dermis (**Figure 4** [figure supplement 1A](#)). Flow cytometry analysis revealed increased infiltration of γ 8 T cells in *Sema4AKO* ear (**Figure 4** [figure supplement 1B](#)). These cells predominantly expressed resident memory T cell (T_{RM})-characteristic molecules, CD69 and CD103 (**Figure 4** [figure supplement 1B](#)). *Sema4AKO* skin also had a higher number of T_{RM} in both CD4 and CD8 T cells (**Figure 4** [figure supplement 1B](#)). The percentages of V γ 2⁺ T cells, DN γ 8 T cells, and CD8 T cells in epidermis were higher in *Sema4AKO* mice than in WT mice, which was not the case in dermis (**Figure 4** [figure supplement 2A](#)). The proportion of V γ 3⁺ dendritic epidermal T cells was comparable between WT mice and *Sema4AKO* mice (**Figure 4** [figure supplement 2A](#)). Epidermal V γ 2⁺ T cells and DN γ 8 T cells in *Sema4AKO* mice showed higher IL-17A-producing capability (**Figure 4D** [figure supplement 2B and C](#)), while IFN γ and IL-4 production was comparable between WT mice and *Sema4AKO* mice (**Figure 4** [figure supplement 2B and C](#)). Conversely, the frequency of IL-17A-producing T cells from dLN was comparable (**Figure 4E** [figure supplement 3](#)). The production of IL-17A and IFN γ from splenic T cells under T17-polarizing conditions remained consistent between WT mice and *Sema4AKO* mice (**Figure 4** [figure supplement 3](#)).

Taken together, it is suggested that T17 cells are specifically upregulated in epidermis, indicating that the epidermal microenvironment plays a pivotal role in facilitating the increased T cell infiltration observed in naïve *Sema4AKO* mice.

Sema4AKO skin shares features with human psoriatic NL

Previous literatures have identified certain features common to psoriatic L and NL, such as thickened epidermis (**Figure 5** [figure supplement 1](#)) (Gallais S  r  zal et al., 2019 [figure supplement 1](#)), CCL20 upregulation (Gallais S  r  zal et al., 2019 [figure supplement 1](#)), and accumulation of IL-17A-producing T cells (Vo et al., 2019 [figure supplement 1](#)), which were detected in *Sema4AKO* mice.

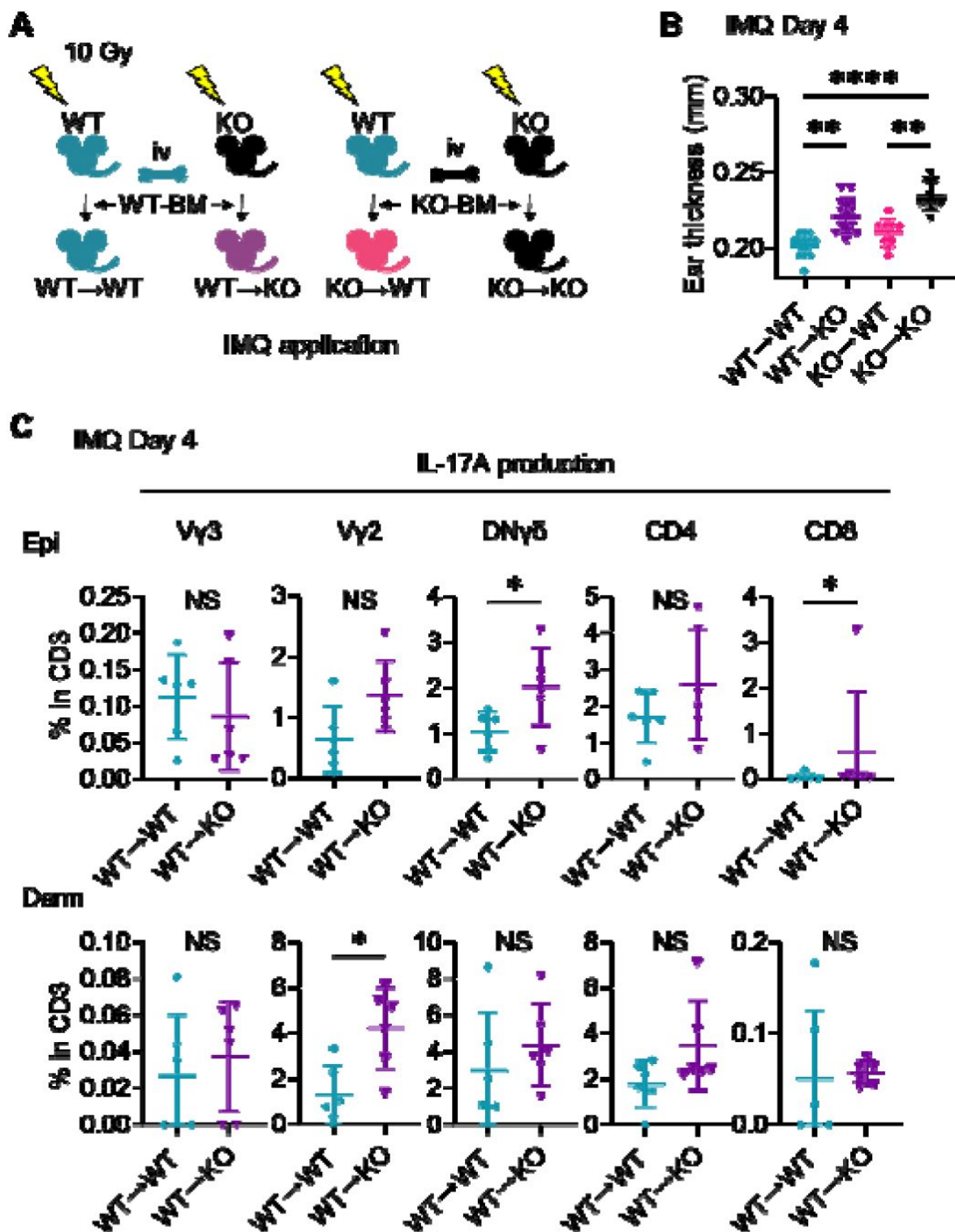


Figure 3

Sema4A in keratinocytes may play a role in preventing murine psoriasis-like dermatitis.

(A) Experimental scheme for establishing BM chimeric mice. (B) IMQ Day 4 ear thickness in the mice with the indicated genotypes ($n = 14$ for WT → WT, $n = 13$ for WT → KO, $n = 9$ for KO → WT, $n = 9$ for KO → KO). (C) The percentages of IL-17A-producing Vγ3, Vγ2, DNγδ, CD4, and CD8 T cells in CD3 fraction from IMQ Day 4 Epi (top) and Derm (bottom) of the ears from WT → WT mice and WT → KO mice ($n = 6$ per group). Each dot represents the average of 4 ear specimens. B-C: * $p < 0.05$, ** $p < 0.01$, **** $p < 0.0001$. NS, not significant. **Figure 3** [source data 1](#) Excel file containing quantitative data for **Figure 3**.

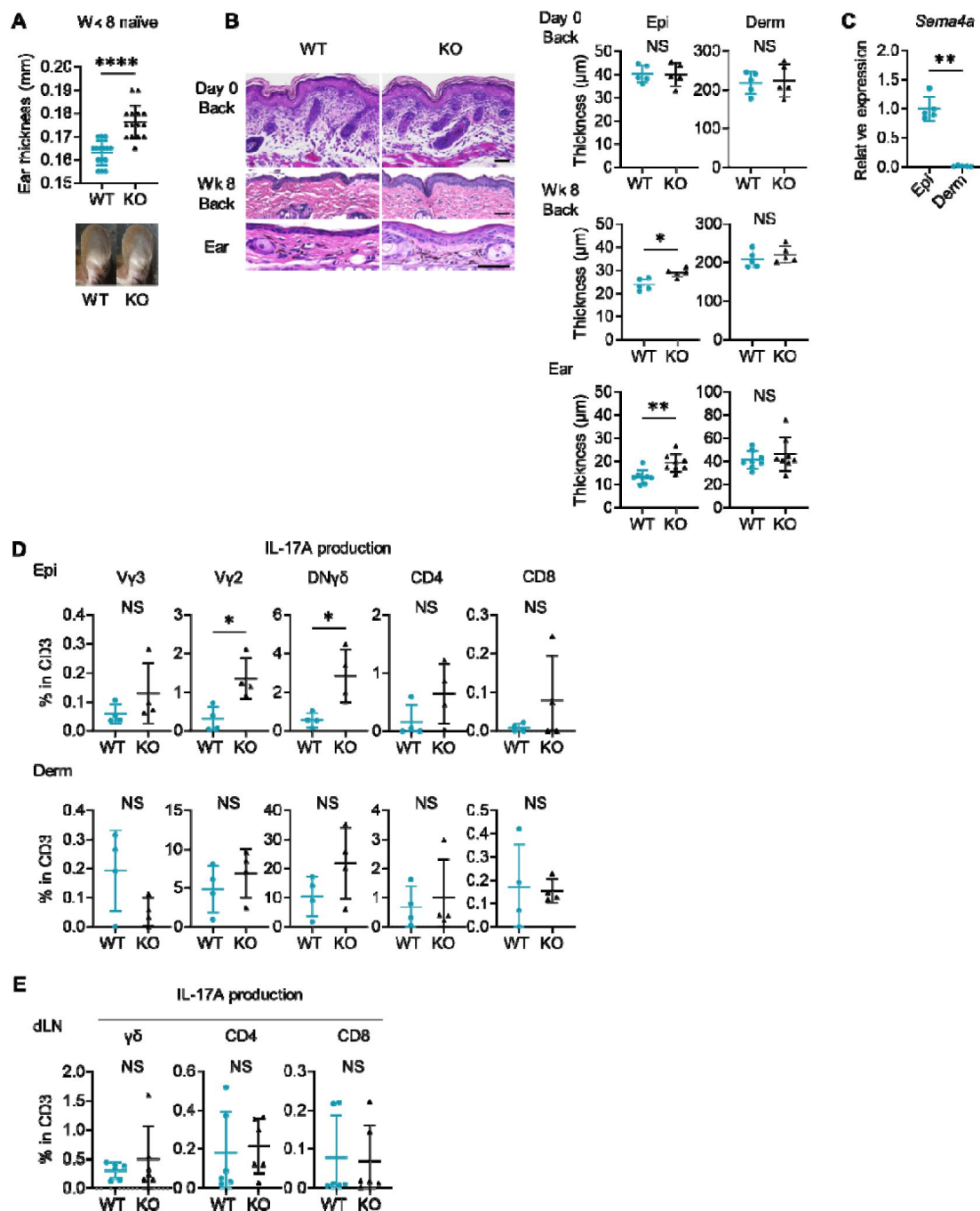


Figure 4

Naïve Sema4AKO epidermis is thicker than WT epidermis with increased $\gamma\delta$ T17 infiltration.

(A) Ear thickness of WT mice and KO mice at week (Wk) 8 ($n = 15$ per group) and representative images. (B) Left: representative Hematoxylin and eosin staining of Day 0 back and Wk 8 back and ear. Scale bar = 50 μm . Right: Epi and Derm thickness in Day 0 back ($n = 5$) and Wk 8 back ($n = 5$) and ear ($n = 8$). (C) Relative *Sema4a* expression in WT Epi and Derm ($n = 5$ per group). (D) The percentages of the IL-17A-producing Vy3, Vy2, DNy δ , CD4, and CD8 T cells in CD3 fraction ($n = 4$ per group) in Epi (top) and Derm (bottom). Each dot represents the average of 4 ear specimens. (E) The graphs showing the percentages IL-17A-producing $\gamma\delta$, CD4, and CD8 T cells in CD3 fraction from draining LN (dLN) of WT mice and Sema4AKO mice ($n = 6$ per group). A-E: * $p < 0.05$, ** $p < 0.01$, **** $p < 0.0001$. NS, not significant. **Figure 4** -source data 1 Excel file containing quantitative data for **Figure 4**.

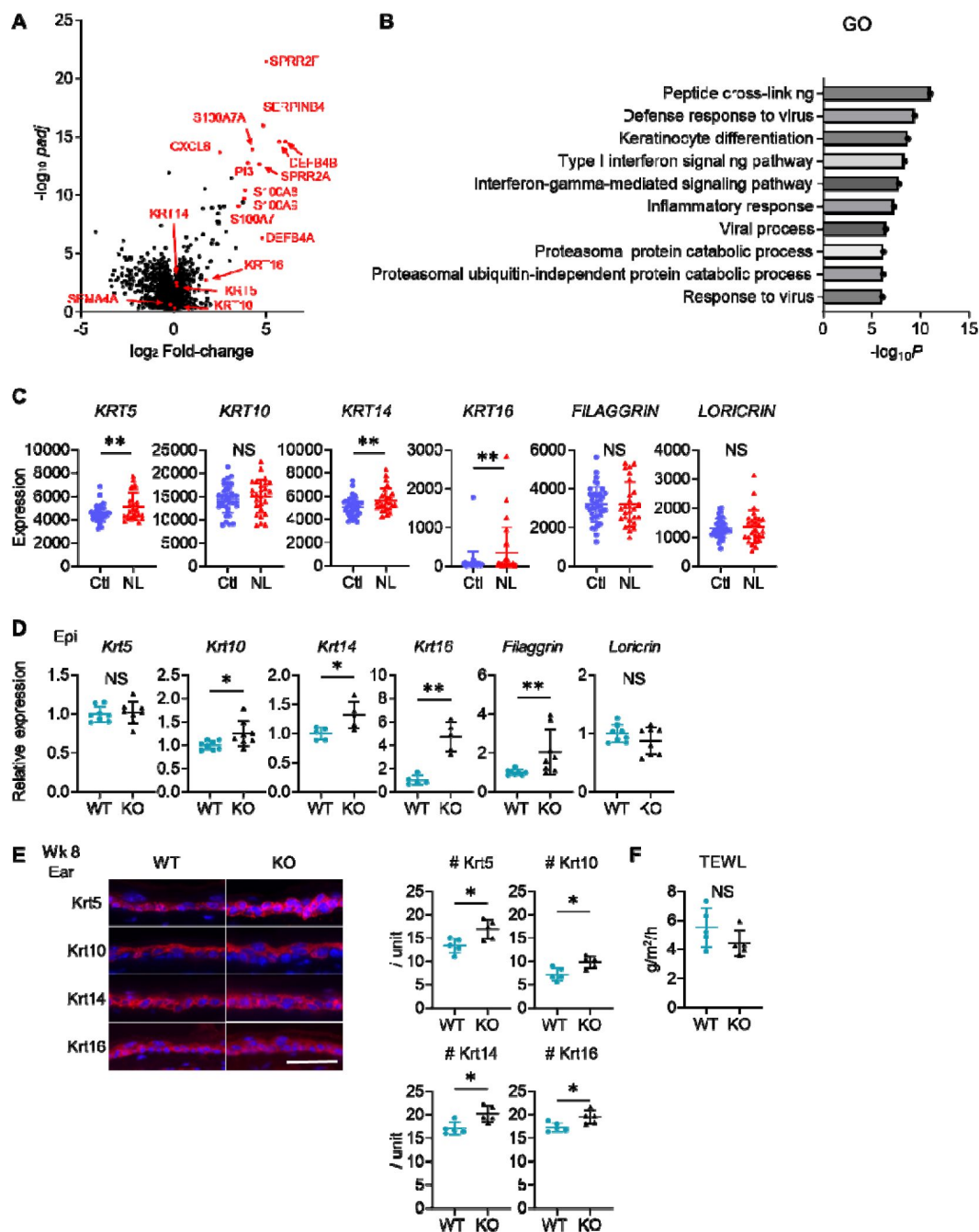


Figure 5

Sema4AKO skin shares the features of human psoriatic NL.

(A and B) The volcano plot (A) and Gene ontology (GO) analysis (B), generated from RNA-sequencing data (GSE121212) using RaNA-seq, display changes in gene expression in psoriatic NL compared to Ctl. (C) The difference in the expression of epidermal differentiation markers between Ctl and NL ($n = 38$ for Ctl, $n = 27$ for NL) was calculated with the transcripts per million values. ** p adj < 0.01. NS, not significant. (D) Relative gene expression of epidermal differentiation markers between wk 8 Epi of WT mice and KO mice ($n = 5$ for Krt14 and Krt16, $n = 8$ for Krt5, Krt10, Filaggrin, and Loricrin). (E) Left: Representative immunofluorescence pictures of Krt5, Krt10, Krt14, and Krt16 (red) overlapped with DAPI. Scale bar = 50 μm. Right: Accumulated graphs showing the numbers of Krt5, Krt10, Krt14, and Krt16 positive cells per 100 μm width ($n = 5$ per group) of wk 8 ear (right). Each dot represents the average from 5 unit areas per sample. (F) Transepidermal water loss (TEWL) in back skin of WT mice and KO mice at wk 8 ($n = 5$ per group). D-F: * p < 0.05, ** p < 0.01. NS, not significant. **Figure 5** [source data](#) 1 Excel file containing quantitative data for **Figure 5** [source data](#).

Gene expression analysis using public RNA-sequencing data (Tsoi et al., 2019) with RaNA-seq (Prieto & Barrios, 2019) showed upregulation of keratinization and antimicrobial peptide genes in NL compared to Ctl (**Figure 5A**). Gene Ontology analysis highlighted an upregulation in biological processes, predominantly in peptide cross-linking involved in epidermis formation and keratinocyte differentiation, with a secondary increase in the defense response to viruses in NL (**Figure 5B**). While the expression of *Keratin (KRT) 10* was comparable between NL and Ctl, upregulation in *KRT5*, *KRT14*, and *KRT16* was observed in NL (**Figure 5C**).

In the murine model, relative expression levels of *Krt10*, *Krt14*, *Krt16*, and *Filaggrin* were elevated in Sema4AKO epidermis. (**Figure 5D**). Immunofluorescence analysis showed that Sema4AKO epidermis had a higher density of keratinocytes positive for Krt5, Krt10, Krt14, and Krt16 compared to WT epidermis (**Figure 5E**). This upregulation was not observed in back skin at birth (**Figure 5**-figure supplement 2). Comparable transepidermal water loss between WT mice and Sema4AKO mice indicated preserved skin barrier function in Sema4AKO mice (**Figure 5F**).

Based on these results, it is implied that the epidermis of human psoriatic NL and Sema4AKO mice exhibit shared pathways, potentially leading to an acanthotic state. Combined with the observed acanthosis and increased T17 infiltration in Sema4AKO mice, Sema4AKO skin is regarded to demonstrate the features characteristic of human psoriatic NL.

mTOR signaling is upregulated in the epidermis of psoriatic NL and Sema4AKO mice

Previous reports have shown that mTOR pathway plays a critical role in maintaining epidermal homeostasis, as evidenced by mice with keratinocyte-specific deficiencies in *Mtor*, *Raptor*, or *Rictor*, which exhibit a hypoplastic epidermis with impaired differentiation and barrier formation (Asrani et al., 2017; Ding et al., 2016; Ding et al., 2020). We thus investigated mTOR pathway in both human and murine epidermis.

Immunohistochemical analyses highlighted the increase in phospho-S6 (p-S6), indicating the upregulation of mTOR complex (C) 1 signaling, in the epidermal upper layers of both L and NL compared to Ctl (**Figure 6A**). The activation of mTORC2 in the epidermis, inferred from phospho-Akt (p-Akt), was scarcely detectable in L, NL, and Ctl (**Figure 6A**). In the epidermis of WT mice and Sema4AKO mice, the upregulation of both mTORC1 and mTORC2 signaling was observed in Sema4AKO epidermis by immunohistochemistry (**Figure 6B**). Western blot analysis showed upregulated mTORC1 signaling in Sema4AKO epidermis compared to WT epidermis (**Figure 6C**). The enhancement of mTORC1 and mTORC2 signaling became obvious in the Sema4AKO epidermis after developing psoriatic dermatitis by imiquimod application (**Figure 6D**).

Inhibition of mTOR signaling modulates cytokeratin expression in Sema4AKO mice

To investigate the contribution of mTORC1 and mTORC2 signaling in the development of psoriatic features in the Sema4AKO epidermis, mTORC1 inhibitor rapamycin and mTORC2 inhibitor JR-AB2-011 were intraperitoneally applied to Sema4AKO mice for 14 days. Although epidermal thickness remained unchanged by the inhibitors (**Figure 7A and B**), relative gene expression of *Krt5* was significantly upregulated and that of *Krt16* was significantly downregulated after rapamycin application (**Figure 7C**). While the upregulation of *Il17a* in Sema4AKO epidermis was not clearly modified by rapamycin (**Figure 7C**), immunofluorescence revealed the decrease in the number of CD3 T cells in Sema4AKO epidermis by rapamycin (**Figure 7D**). We additionally conducted topical application of rapamycin gel and vehicle gel on the left and right ears of Sema4AKO mice, respectively. Although there were no detectable changes in epidermal thickness and epidermal T

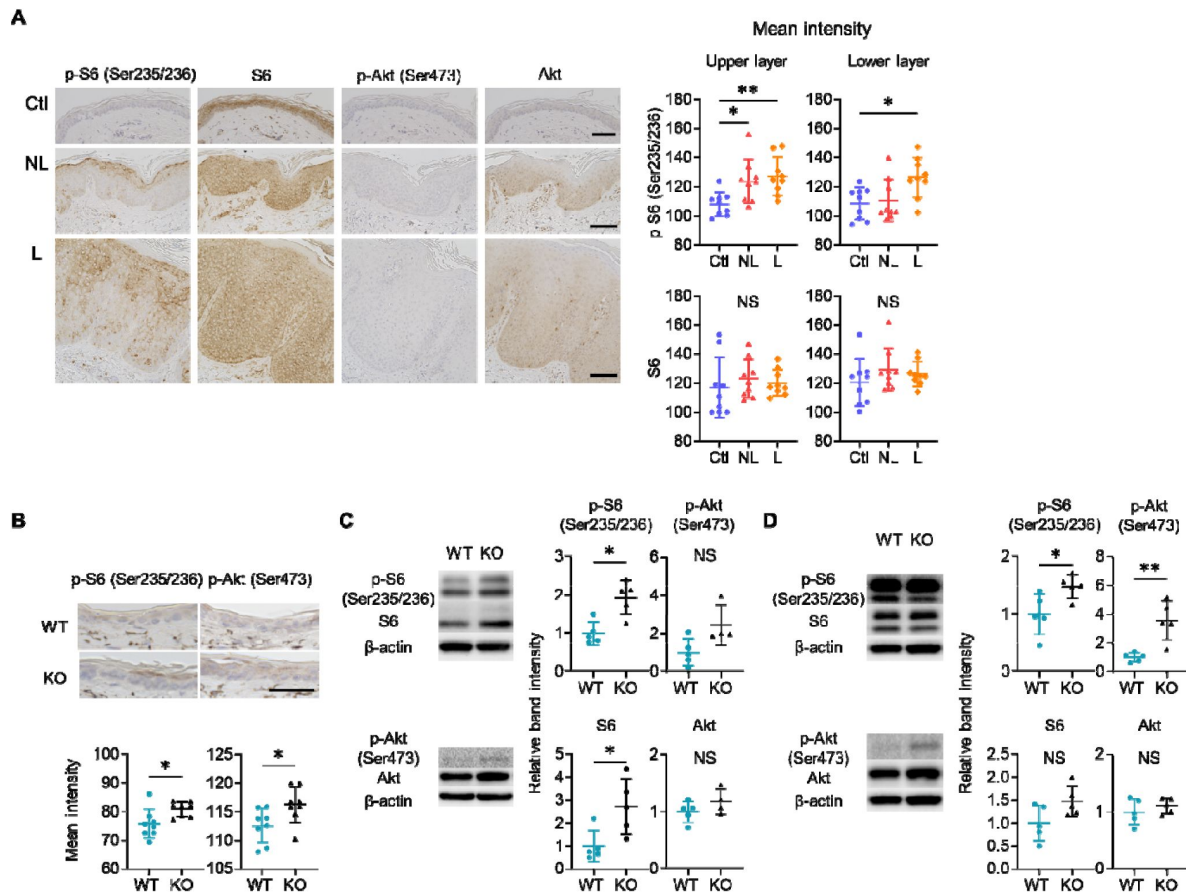


Figure 6

mTOR signaling is upregulated in the epidermis of psoriatic NL and Sema4AKO mice.

(A) Representative results of immunohistochemistry displaying cells positive for p-S6 (Ser235/236), S6, p-Akt (Ser473), and Akt in Ctl, NL, and L. The graphs of accumulated data show the mean intensity of p-S6 and S6 in the upper and lower epidermal layers ($n = 9$ per group). Scale bar = 100 μm . Each dot represents the average mean intensity from 5 unit areas per sample. (B) The mean intensity of p-S6 (Ser235/236) and p-Akt (Ser473), detected by immunohistochemistry in the epidermis of WT mice and KO mice, were analyzed. Scale bar = 50 μm . Each dot represents the average intensity from 5 unit areas per sample ($n = 8$ per group). (C and D) Immunoblotting of p-S6 (Ser235/236), S6, p-Akt (Ser473), and Akt in tissue lysates from epidermis without treatment (C) and with IMQ treatment for consecutive 4 days (D) ($n = 5$ per group, except for p-Akt and Akt in C, for which $n = 4$). A-D: $*p < 0.05$, $**p < 0.01$. NS, not significant. Figure 6-source data 1 Excel file containing quantitative data for Figure 6.

cell counts, the upregulation of *Krt5* and downregulation of *Krt16* was observed again (**Figure 7** [figure supplement 1](#)). Conversely, the application of JR-AB2-011 resulted in decreased expression of *Krt5*, *Krt10*, and *Krt14* with a trend towards increased *Krt16* expression (**Figure 7E** [figure supplement 1](#)). JR-AB2-011 did not influence the number of infiltrating T cells in the epidermis (**Figure 7F** [figure supplement 1](#)). Next, we investigated whether intraperitoneal rapamycin treatment effectively downregulates inflammation in the IMQ-induced murine model of psoriasis in *Sema4AKO* mice (**Figure 7** [figure supplement 2A](#)). Rapamycin significantly reduced epidermal thickness compared to vehicle treatment (**Figure 7** [figure supplement 2B](#)). Additionally, rapamycin treatment downregulated the expression of *Krt10*, *Krt14*, and *Krt16* (**Figure 7** [figure supplement 2C](#)). While the upregulation of *Il17a* in the *Sema4AKO* epidermis in IMQ model was not clearly modified by rapamycin (**Figure 7** [figure supplement 2C](#)), immunofluorescence revealed a decrease in the number of CD3 T cells in *Sema4AKO* epidermis by rapamycin (**Figure 7** [figure supplement 2D](#)). In the naïve states, mTORC1 primarily regulates keratinocyte proliferation, whereas mTORC2 mainly involved in the keratinocyte differentiation through *Sema4A*-related signaling pathways. Conversely, in the psoriatic dermatitis state, rapamycin downregulated both keratinocyte differentiation and proliferation markers. The observed similarities in *Il17a* expression following treatment with rapamycin and JR-AB2-011, regardless of additional IMQ treatment, suggest that *Il17a* production is not significantly dependent on *Sema4A*-related mTOR signaling.

Discussion

Recent studies have highlighted the significance of IL-17A-producing T_{RM} in psoriatic NL (Cheuk et al., 2014 [figure supplement 1](#); Gallais Sérézal et al., 2018 [figure supplement 1](#); Vo et al., 2019 [figure supplement 1](#); Vu, Koguchi-Yoshioka, & Watanabe, 2021 [figure supplement 1](#)). This condition is also characterized by epidermal thickening with elevated CCL20 expression (Gallais Sérézal et al., 2019 [figure supplement 1](#)) and a distinct cytokeratin expression pattern, marked by increased levels of *Krt5*, *Krt14*, and *Krt16* (Tsoi et al., 2019 [figure supplement 1](#)) (**Figure 5C** [figure supplement 1](#)). Here, we identified that murine *Sema4A* deficiency could induce these features of psoriatic NL.

Sema4A expression in epidermis, but not in dermis, was diminished in psoriatic L and NL compared to Ctl. While *Sema4A* expression on blood immune cells was upregulated in psoriasis, its serum levels were comparable between Ctl and psoriasis. Despite the reported involvement of increased *Sema4A* expression on immune cells in the pathogenesis of multiple sclerosis (Koda et al., 2020 [figure supplement 1](#); Nakatsuji et al., 2012 [figure supplement 1](#)), our results suggest that the diminished expression of *Sema4A* in skin-constructing cells plays a more prominent role in the pathogenesis of psoriasis than its increased expression on immune cells.

In our murine model, regardless of imiquimod application, IL-17A-producing T cells were increased in *Sema4AKO* skin although their frequency was comparable in dLN of WT mice and *Sema4AKO* mice. Additionally, the potential for *in vitro* T17 differentiation did not differ between T cells from WT mice and *Sema4AKO* mice. These findings suggest that the absence of *Sema4A* in the skin microenvironment plays a crucial role in the localized upregulation of IL-17A-producing T cells in *Sema4AKO* mice.

It is well-documented that upregulated mTORC1 signaling promotes the pathogenesis of psoriasis (Buerger, 2018 [figure supplement 1](#); Karagianni, Pavlidis, Malakou, Piperi, & Papadavid, 2022 [figure supplement 1](#); Ruf, Andreoli, Itin, Pluschke, & Schmid, 2014 [figure supplement 1](#)), and that rapamycin can partially ameliorate the disease activity in psoriatic subjects and murine models (Bürger et al., 2017 [figure supplement 1](#); Gao & Si, 2018 [figure supplement 1](#); Reitamo et al., 2001 [figure supplement 1](#)). Our results using *Sema4AKO* mice revealed that inhibition of mTORC1 leads to the downregulation of *Krt16*, and inhibition of mTORC2 leads to the downregulation of undifferentiated keratinocytes, while the *Sema4AKO* epidermal thickening and the upregulated IL-17A signaling are not reversed by mTOR blockade in the naïve state. It is plausible that the

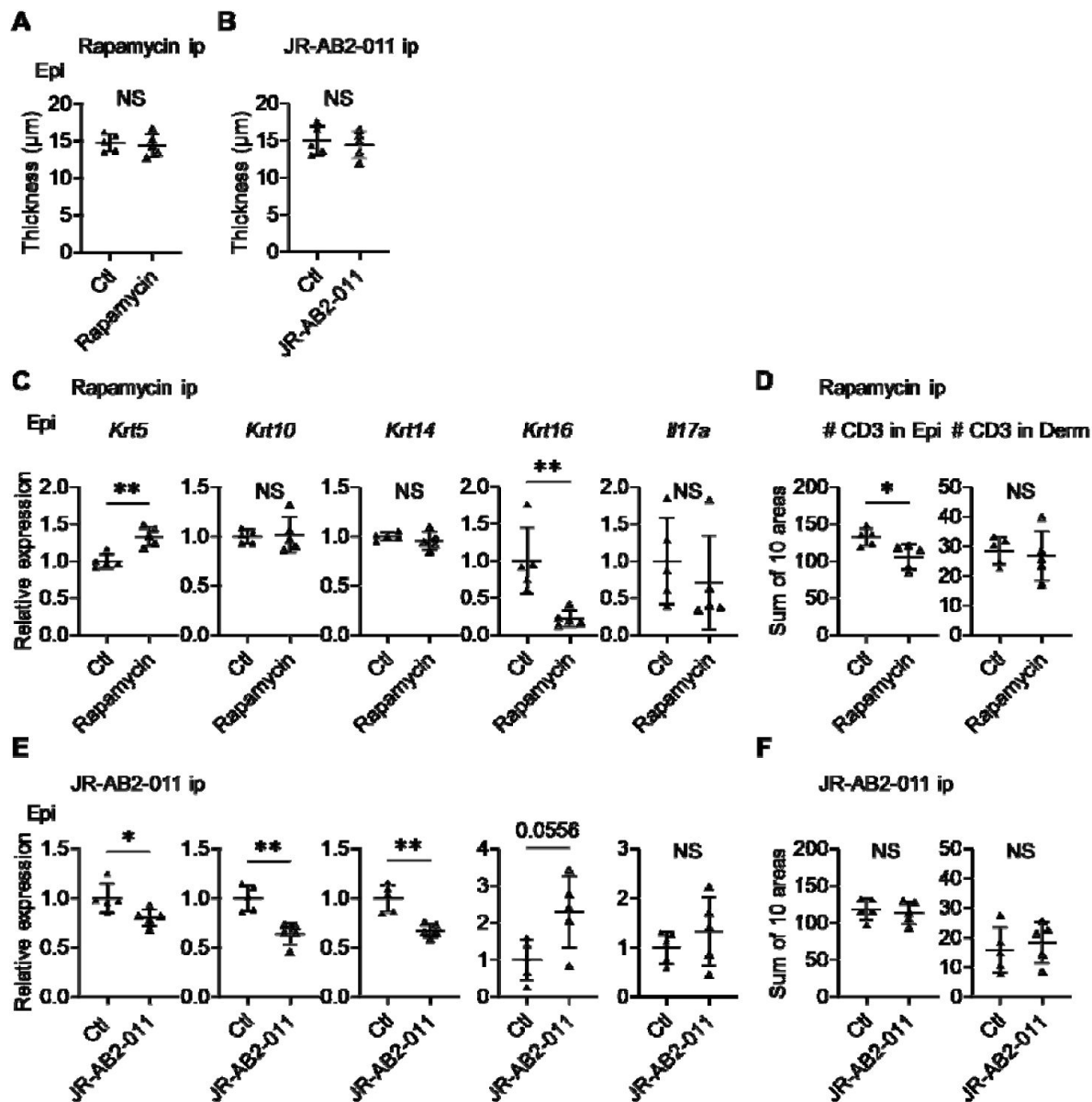


Figure 7

Inhibitors of mTOR signaling modulate the expression of cytokeratinins in Sema4AKO mice.

(A and B) Epidermal thickness of Sema4AKO mice treated intraperitoneally with vehicle (Ctl) or rapamycin (A), and Ctl or JR-AB2-011 (B) ($n = 5$ per group). (C and D) Relative expression of keratinocyte differentiation markers and *I17a* in Sema4AKO Epi (C), and the number of T cells in Epi and Derm under Ctl or rapamycin (D) ($n = 5$ per group). (E and F) Relative expression of keratinocyte differentiation markers and *I17a* in Sema4AKO Epi (E), and the number of T cells in Epi and Derm under Ctl or JR-AB2-011 (F) ($n = 5$ per group). D and F: Each dot represents the sum of numbers from 10 unit areas across 3 specimens. A-F: * $p < 0.05$, ** $p < 0.01$. NS, not significant. Figure 7-source data 1 Excel file containing quantitative data for Figure 7.

downregulation of *Sema4A* can lead to the upregulation of mTORC1 and mTORC2 signaling in keratinocytes, and the augmented signaling leads to the psoriatic profile of proliferation and differentiation of keratinocytes, which is part of the psoriatic NL disposition.

This study has limitations. *Sema4A* expression in skin cells other than keratinocytes was not thoroughly investigated. However, since single-cell RNA-sequencing has shown that *Sema4A* is predominantly expressed in keratinocytes, dendritic cells, and macrophages, with a notable reduction in keratinocytes in psoriasis, we infer that keratinocytes are the primary cells responsible for the psoriatic features resulting from *Sema4A* downregulation. We were not able to determine whether *Sema4A* functions as a ligand or a receptor in epidermis (Ito & Kumanogoh, 2016 [DOI](#); Kumanogoh et al., 2002 [DOI](#); Lu et al., 2018 [DOI](#); Sun et al., 2017 [DOI](#)) in this study, either. We were not able to reveal how *Sema4A* expression is regulated. Although we showed that downregulation of *Sema4A* is related to the abnormal cytokeratin expression observed in psoriasis, we could not determine the relationships between *Sema4A* expression and the essential molecules upregulated in psoriatic keratinocytes. While both mTORC1 and mTORC2 signals are upregulated in *Sema4AKO* epidermis, we were not able to confirm mTORC2 signaling from human skin due to technical limitation and sample limitation, while the results from augmented mTORC2 signal in *Sema4AKO* mice and the normalization of *Krt5* and *Krt14* by mTORC2 inhibitor imply the involvement of mTORC2 signal in psoriatic epidermis. The role of *Sema4A* other than mTOR signaling, which may be involved in the regulation of T17 induction in skin, is not discussed, either. These limitations should be overcome in the near future.

In summary, epidermal *Sema4A* downregulation can reflect the psoriatic non-lesional features. Thus, targeting the downregulated *Sema4A* and upregulated mTOR signaling in psoriatic epidermis can be a promising strategy for psoriasis therapy and modification of psoriatic diathesis in NL for the prevention of disease development.

Materials and methods

Human sample collection

Psoriatic L and NL skin specimens were acquired from 17 psoriasis patients. Ctl specimens were obtained from 19 subjects who underwent tumor resection or reconstructive surgery. For epidermal-dermal separation, specimens were incubated overnight at 4°C with 2.5 mg/mL dispase II (Wako, Osaka, Japan) in IMDM (Wako). Blood samples were collected from 73 psoriasis and 33 Ctl. In addition to ELISA (Nakatsuji et al., 2012 [DOI](#)), peripheral blood mononuclear cells were isolated using Ficoll-Paque PLUS density gradient media (Cytiva, Tokyo, Japan). Patient details are provided in Supplemental Table 1.

Mice

C57BL/6J WT mice were procured from CLEA Japan (Tokyo, Japan). *Sema4AKO* mice with C57BL/6J background were generated as previously described (Kumanogoh et al., 2005 [DOI](#)). We examined female mice in order to reduce the result variation. Neonatal mice and female mice aged 8 to 12 weeks, maintained under specific pathogen-free conditions, were used.

To develop psoriasis-like dermatitis, 10 mg of 5% imiquimod cream (Mochida, Tokyo, Japan) was applied to both ears for 4 consecutive days. To induce IL-23-mediated psoriasis-like dermatitis, 20 µl of phosphate-buffered saline containing 500 ng of recombinant mouse IL-23 (BioLegend, San Diego, CA) was injected intradermally into both ears of anesthetized mice using a 29-gauge needle for 4 consecutive days. Ear thickness was measured using a thickness gauge (Peacock, Tokyo, Japan).

For bone marrow reconstitution, 2×10^5 bone marrow cells obtained from the indicated strains were intravenously transferred to recipient mice that had received a single 10 Gy irradiation. The reconstituted mice were subjected to the experiments in 8 weeks.

In the specified experiments, skin specimens were separated into epidermis and dermis by 5 mg/mL dispase (Wako) in IMDM for 30 minutes at 37°C.

Transepidermal water loss measurements were performed on the back skin of mice at week 8 using a Tewameter (Courage and Khazaka Electronic GmbH, Cologne, Germany), according to the manufacturer's instructions.

Immunofluorescence and immunohistochemistry

Specimens were fixed in 4% paraformaldehyde phosphate buffer solution (Wako), embedded in paraffin, and sliced into 3- μ m thickness on glass slides. After deparaffinization and rehydration, antigen retrieval was performed using citrate buffer (pH 6.0, Nacalai tesque, Kyoto, Japan) or TE buffer (pH 9.0, Nacalai tesque).

For immunohistochemistry, samples were incubated with 3% H₂O₂ (Wako) for 5 minutes. After blocking (Agilent, Santa Clara, CA), the specimens were incubated with the indicated primary antibodies under specified conditions (Supplemental Table 2). The specimens were applied with the Dako REAL EnVision Detection System, Peroxidase/DAB, Rabbit/Mouse, HRP kit (Agilent) and counterstained with hematoxylin (Wako). For immunofluorescence, the blocked specimens were incubated with the indicated primary antibodies followed by the secondary antibodies (Table E4). Mounting medium with DAPI (Vector Laboratories, Burlingame, CA) was used. Slides were observed using a fluorescence microscope (BX-X700, Keyence, Osaka, Japan). The staining intensity was measured using Fiji software (ImageJ, National Institutes of Health, Bethesda, MD) over lengths of 200 μ m in human samples and 100 μ m in murine samples. This measurement was taken from five areas of each specimen, and the average score is presented. The average thickness of epidermis and dermis from 10 spots is presented. The number of epidermal cells positive for the indicated cytokeratins was counted per 100 μ m width, with the average number from 5 areas being presented. Additionally, the numbers of CD3-positive cells in murine ears were counted across 10 fields of 400 μ m width, with the total sum being presented.

Quantitative reverse transcription polymerase chain reaction (qRT-PCR)

Total RNA was extracted from homogenized skin tissue using Direct-zol RNA MiniPrep Kit (ZYMO Research, Irvine, CA). cDNA was synthesized by High-Capacity RNA-to-cDNATM Kit (Thermo Fisher Scientific, Waltham, MA). qPCR was performed using TB Green Premix Ex TaqTM II (Takara-bio, Shiga, Japan) on a ViiA 7 Real-Time PCR System (Thermo Fisher Scientific). The primers are listed in Supplemental Table 3. All samples were run in triplicate, and the median CT value was calculated. Relative gene expression levels were normalized to the housekeeping gene *GAPDH* using the $\Delta\Delta$ CT technique.

Flow cytometry analysis

Single-cell suspensions from murine skin-dLN and spleen were prepared by grinding, filtering, and lysing red blood cells (BioLegend). Skin specimens were minced and digested with 3 mg/mL collagenase type III (Worthington Biochemical Corporation, Lakewood, NJ) in RPMI 1640 medium (Wako) at 37°C for 10 minutes for epidermis, and 30 minutes for dermis or whole skin. Cells were surface-stained with directly conjugated monoclonal antibodies (Supplemental Table 4). Dead cells were identified using LIVE/DEADTM Fixable Dead Cell Stain Kit (Thermo Fisher Scientific). To evaluate cytokine production, cells were stimulated with Phorbol 12-Myristate 13-Acetate (PMA; 50 ng/mL, Wako) and ionomycin (1000 ng/mL, Wako), plus BD GolgiplugTM (BD Biosciences, San

Jose, CA) for 4 hours before surface staining. Fixation, permeabilization, and intracellular cytokine staining were performed using BD Cytofix/Cytoperm™ Fixation/Permeabilization Kit (BD Biosciences) according to the manufacturer's protocol. In specified experiments, the numbers of each cell subset per ear were estimated using CountBright™ Absolute Counting Beads (Thermo Fisher Scientific). Sample analysis was conducted using BD FACSCanto II (BD Biosciences), and data were analyzed with Kaluza software (Beckman Coulter, Brea, CA).

***In vitro* Th17 differentiation**

Murine splenic T cells were isolated using Pan T Cell Isolation Kit II (Miltenyi Biotec, Bergisch Gladbach, Germany). Two hundred thousand T cells per well were cultured in 96-well plates in the presence of T Cell Activation/Expansion Kit (Miltenyi Biotec) for 2 weeks. The medium was supplemented twice per week with the following recombinant cytokines: mouse recombinant IL-6 (20[ng/mL], IL-1 β (20[ng/mL], and IL-23 (40[ng/mL] for the IL-23 dependent Th17 cell condition; IL-6 (20[ng/mL] and TGF β 1 (3[ng/mL] for the IL-23 independent Th17 cell condition; or IL-23 (40[ng/mL] for the IL-23 only Th17 cell condition. The cytokines listed were purchased from BioLegend (Supplemental Table 5). Afterward, the cultured cells were processed for flow cytometry analysis.

Western blotting

Murine epidermis was lysed using RIPA buffer (Wako) containing phosphatase-and protease-inhibitor cocktail (Nacalai tesque). Protein lysates were separated by 10% SuperSep™ Ace (Wako), transferred onto polyvinylidene difluoride membranes (0.45 μ m, Merck, Darmstadt, Germany) by Trans-Blot Turbo Transfer System (Bio-Rad, Hercules, CA). The membrane was blocked with 5% bovine serum albumin and subjected to immunoblotting targeting the indicated proteins overnight 4°C, followed by the application of HRP-conjugated secondary antibody for one hour at room temperature (Supplemental Table 2). WB stripping solution (Nacalai tesque) was used to remove the antibodies for further evaluation.

Inhibition of mTOR

Rapamycin (4 mg/kg, Sanxin Chempharma, Hebei, China), JR-AB2-011 (400 μ g/kg, MedChemExpress, Monmouth Junction, NJ) and vehicle were applied intraperitoneally to mice once daily for 14 consecutive days. Rapamycin and JR-AB2-011 were dissolved in 100% DMSO and diluted with 40% Polyethylene Glycol 300 (Wako), 5% Tween-80 (Sigma-Aldrich, St. Louis, MO) and 45% saline in sequence. For topical application, 0.2% rapamycin gel and vehicle gel were prepared by the pharmaceutical department as previously described (Wataya-Kaneda et al., 2017 [\[17\]](#)). Rapamycin gel was applied to the left ear, and vehicle gel to the right ear (10mg/ear) of Sema4AKO, once daily for 14 consecutive days. To analyze the preventive effectiveness of rapamycin in an IMQ-induced murine model of psoriatic dermatitis, Sema4AKO mice were administered either vehicle or rapamycin intraperitoneally from Day 0 to Day 17, and IMQ was topically applied to both ears for 4 days starting on Day 14. Then, on Day 18, ears were collected for further analysis.

Data processing of single-cell RNA-sequencing and bulk RNA-sequencing

The raw count matrix data from the previously reported single-cell RNA-sequencing data from GSE220116 (Kim et al., 2023 [\[18\]](#)) were imported into Scanpy (1.9.6) using Python for further analyses. For each sample, cells and genes meeting the following criteria were excluded: cells expressing over 200 genes (sc.pp.filter_cells), cells with a high proportion of mitochondrial genes (> 5%), and genes expressed in fewer than 3 cells (sc.pp.filter_genes). Counts were normalized using sc.pp.normalize_per_cell, logarithmized (sc.pp.log1p), and scaled (sc.pp.scale). Highly variable genes were selected using sc.pp.filter_genes_dispersion with the options min_mean = 0.0125, max_mean = 2.5, and min_disp = 0.7.

Principal Component analysis was conducted with `sc.pp.pca`, selecting the 1st to 50th Principal Components for embedding and clustering. Neighbors were calculated with batch effect correction by BBKNN (Polański et al., 2020 [DOI](#)). Embedding was performed by `sc.tl.umap`, and cells were clustered using `sc.tl.leiden`. Some cells were further subclassified in a similar manner. The data was integrated into an `h5ad` file, which can be visualized in Cellxgene VIP (K. Li et al., 2022 [DOI](#)). We then performed differential analysis between two groups of cells to identify differential expressed genes using Welch's t-test. Multiple comparisons were controlled using the Benjamini-Hochberg procedure, with the false discovery rate set at 0.05 and significance defined as $padj < 0.05$.

Bulk RNA-sequencing data from GSE121212 (Tsoi et al., 2019 [DOI](#)) were re-analyzed using RaNAseq (Prieto & Barrios, 2019 [DOI](#)) for differential expression in Ctl versus psoriatic NL. Our analysis included normalization, differential gene expression (defining significance as $padj < 0.05$), and focused on Gene Ontology biological process analysis. The gene expression was calculated with the transcripts per million values.

Statistical analysis

GraphPad Prism 10 software (GraphPad Software, La Jolla, CA) was used for all statistical analyses except for RNA-sequencing. Mann-Whitney test was used for two-group comparisons, while Kruskal-Wallis test followed by Dunn's multiple comparisons test was applied for comparison among three or more groups. Statistical significance was defined as $p < 0.05$ (*), $p < 0.01$ (**), and $p < 0.001$ (***). In the analysis of RNA-sequencing data, statistical significance was defined as $padj < 0.01$ (**), and $padj < 0.001$ (***).

Study approval

All experiments involving human specimens were in accordance with the Declaration of Helsinki and were approved by the Institutional Review Board in Osaka University Hospital (20158-6). Written informed consent was obtained from all participants. All murine experiments were approved by Osaka University Animal Experiment Committee (J007591-013) and all procedures were conducted in compliance with the Guidelines for Animal Experimentation established by Japanese Association for Laboratory Animal Science.

Data availability

The single-cell RNA-sequencing datasets generated by Kim et al. (Kim et al., 2023 [DOI](#)) and Tsoi et al. (Tsoi et al., 2019 [DOI](#)) used in this study are available in the NCBI Gene Expression Omnibus under accession codes GSE220116 and GSE121212, respectively. Values for all data points in graphs are reported in the Source data.

Acknowledgements

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Additional information

Competing interests

SN and SM are affiliated with Maruho Co. as employees, but have declared no conflicts of interest related to this research. The remaining authors also declare no conflicts of interest.

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Supplementary materials

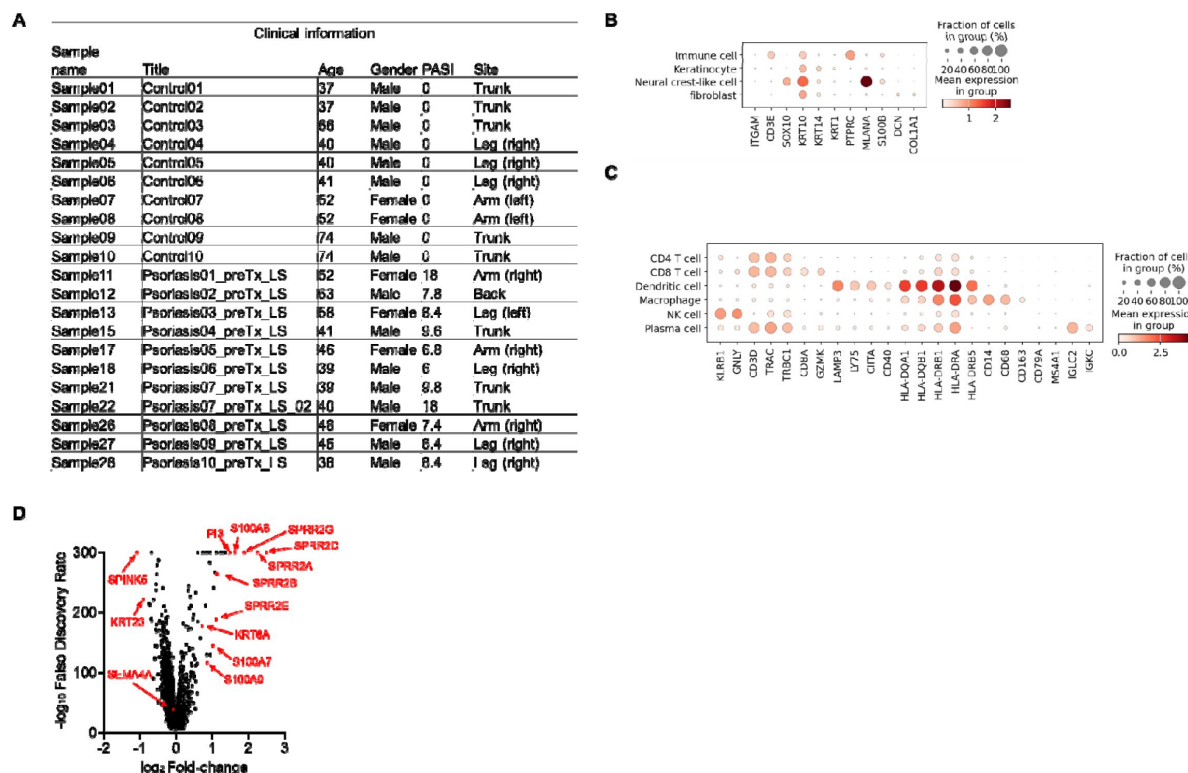


Figure 1-figure supplement 1

Sema4A is downregulated in the keratinocytes of lesional psoriasis in the single-cell RNA-sequencing data.

(A) Sample information for specimens from Ctl and psoriatic L (GSE220116). (B and C) Clusters of cells were identified by their expression patterns of signature genes. (D) The volcano plot displays changes in gene expression in psoriatic L compared to Ctl. Figure 1-figure supplement 1-source data 1 Excel file containing quantitative data for Figure 1-figure supplement 1.

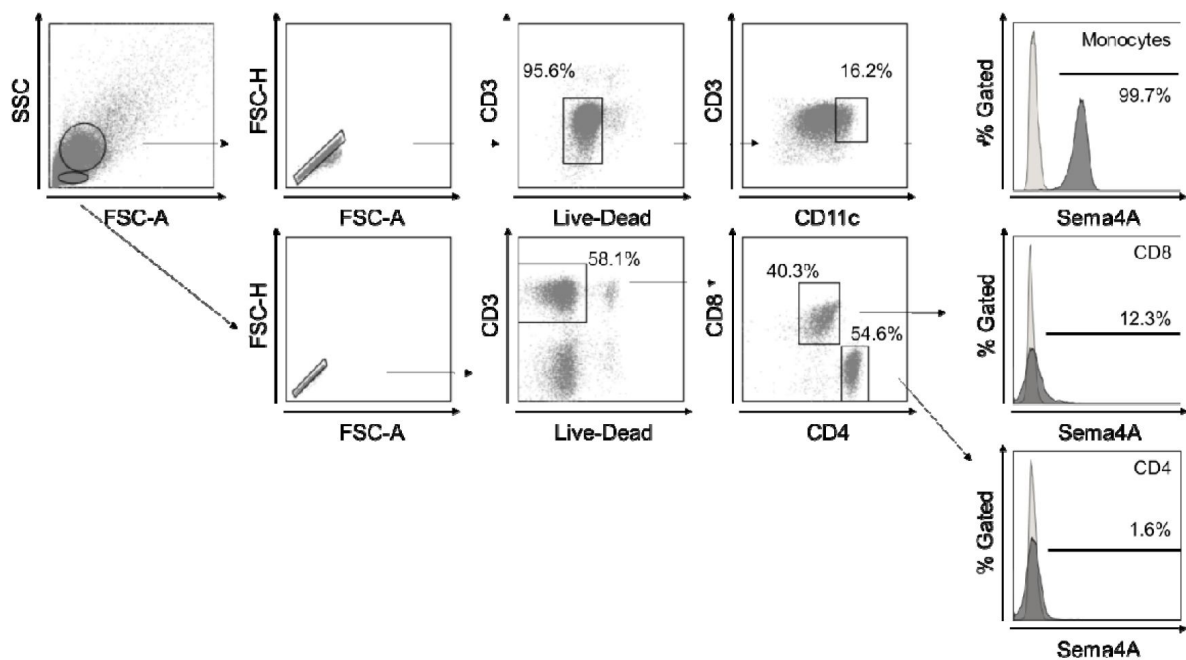


Figure 1-figure supplement 2

Gating strategy in flow cytometry.

Gating strategy for human Sema4A expression in blood cells. Large and small cells were distinguished using forward scatter (FCS) and side scatter (SSC) in a dot plot panel, with dead cells being excluded. Monocytes were defined within the live large cell population as CD11c positive. CD4 and CD8 T cells were identified within the live small cell population as CD3 positive CD4 positive and CD3 positive CD8 positive populations, respectively. The empty histogram represents the flow cytometry minus one control for Sema4A.

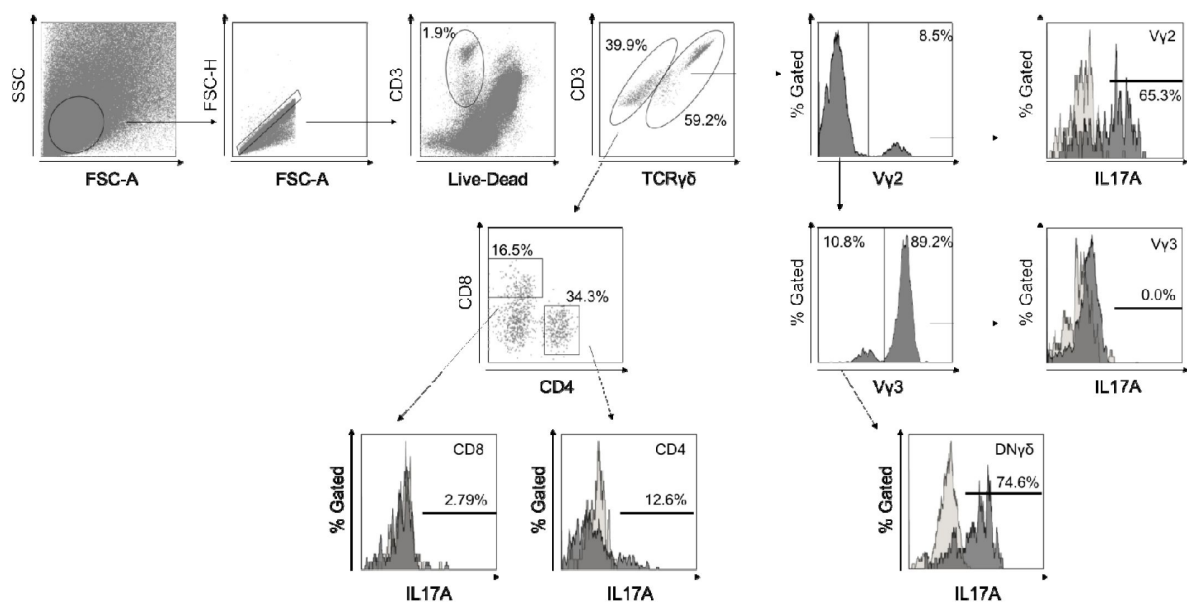


Figure 2-figure supplement 1

Gating strategy in flow cytometry.

Gating strategy for murine T cells infiltrating the epidermis and dermis. After excluding dead cells, TCRγδ positive T cells were evaluated for the expression of Vγ2. TCRγδ positive Vγ2 negative population was further assessed the expression of Vγ3. The CD3 positive TCRγδ negative population was evaluated for the expression of CD4 and CD8. Each population was analyzed for cytokine production. The empty histogram represents the isotype control for IL-17A.

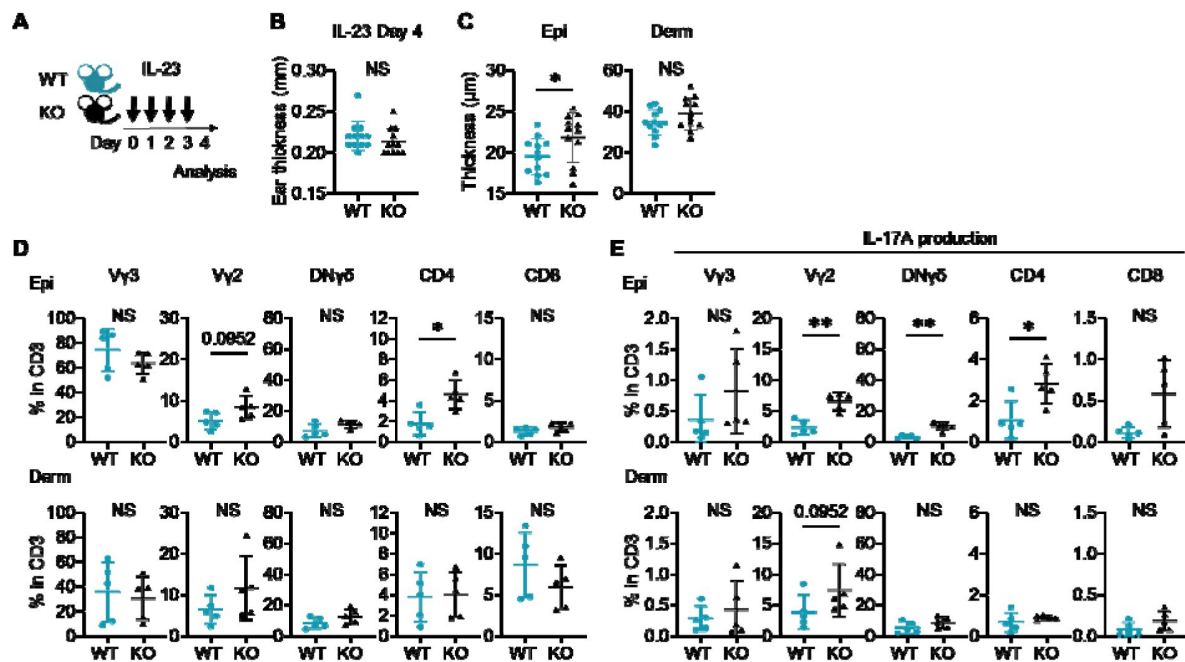


Figure 2-figure supplement 2

IL-23-mediated psoriasis-like dermatitis is augmented in Sema4AKO mice.

(A) An experimental scheme involved intradermally injecting 20 μ l of phosphate-buffered saline containing 500 ng of recombinant mouse IL-23 into both ears of WT mice and KO mice for 4 consecutive days. Samples for following analysis were collected on Day 4. (B and C) Ear thickness (B) and Epi and Derm thickness (C) of WT mice and KO mice on Day 4 ($n = 12$ per group). (D and E) The percentages of Vy3, Vy2, DNy5, CD4, and CD8 T cells (D) and those with IL-17A production (E) in CD3 fraction in the Epi (top) and Derm (bottom) of WT and KO ears ($n = 5$ per group). Each dot represents the average of 4 ear specimens. B-E: * $p < 0.05$, ** $p < 0.01$. NS, not significant. Figure 2-figure supplement 2-source data 1 Excel file containing quantitative data for Figure 2-figure supplement 2.

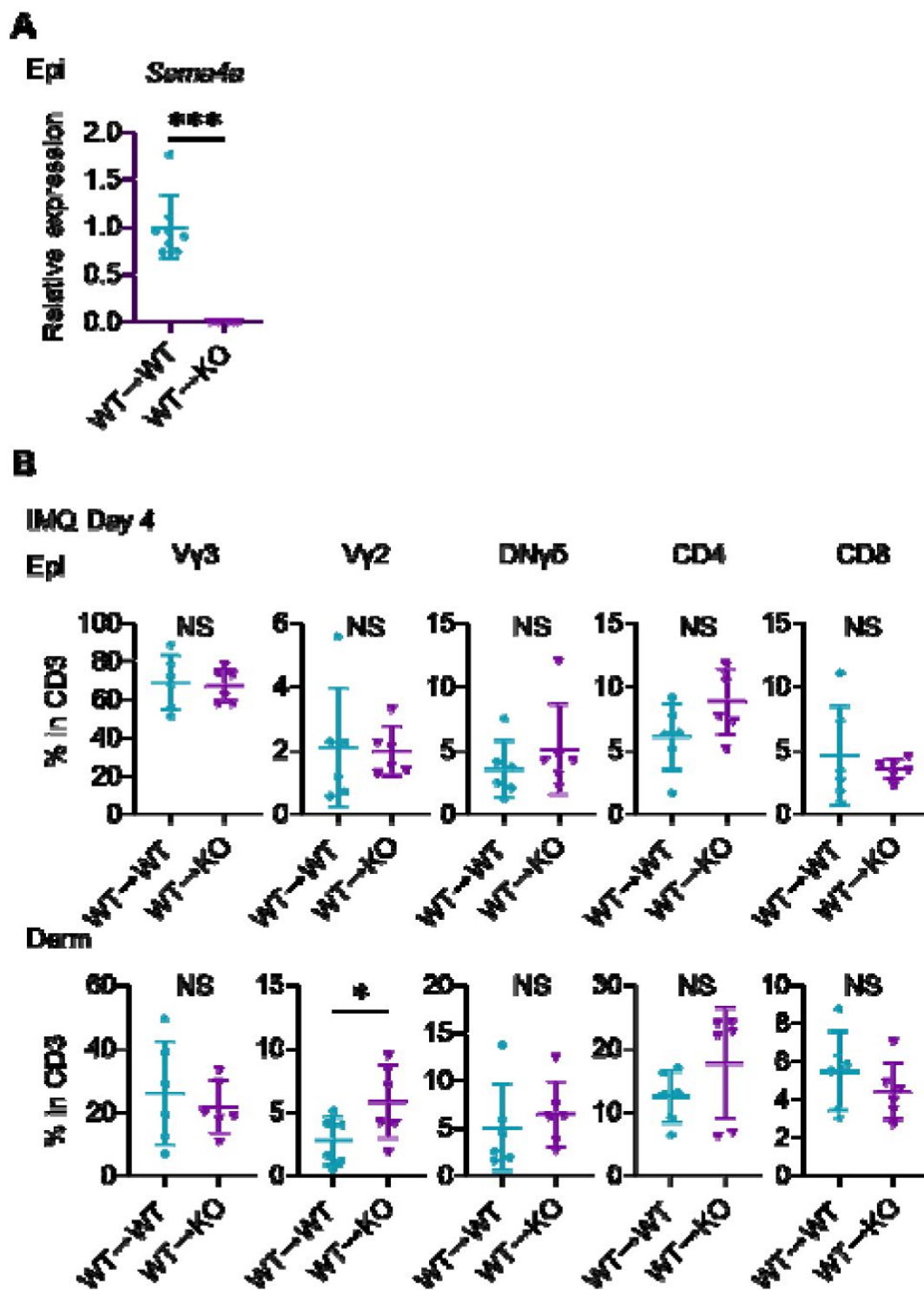


Figure 3-figure supplement 1

T cells' fractions infiltrating in the chimeric mice ear.

(A) *Sema4a* expression in the Epi of WT → WT mice and WT → KO mice ($n = 8$ for WT → WT, $n = 7$ for WT → KO). (B) The percentages of Vy3, Vy2, DNYδ, CD4, and CD8 T cells in CD3 fraction from IMQ Day 4 Epi (top) and Derm (bottom) of the ears from WT → WT mice and WT → KO mice ($n = 6$ per group). Each dot represents the average of 4 ear specimens. A-B: * $p < 0.05$, *** $p < 0.001$. NS, not significant. Figure 3-figure supplement 1-source data 1 Excel file containing quantitative data for Figure 3-figure supplement 1.

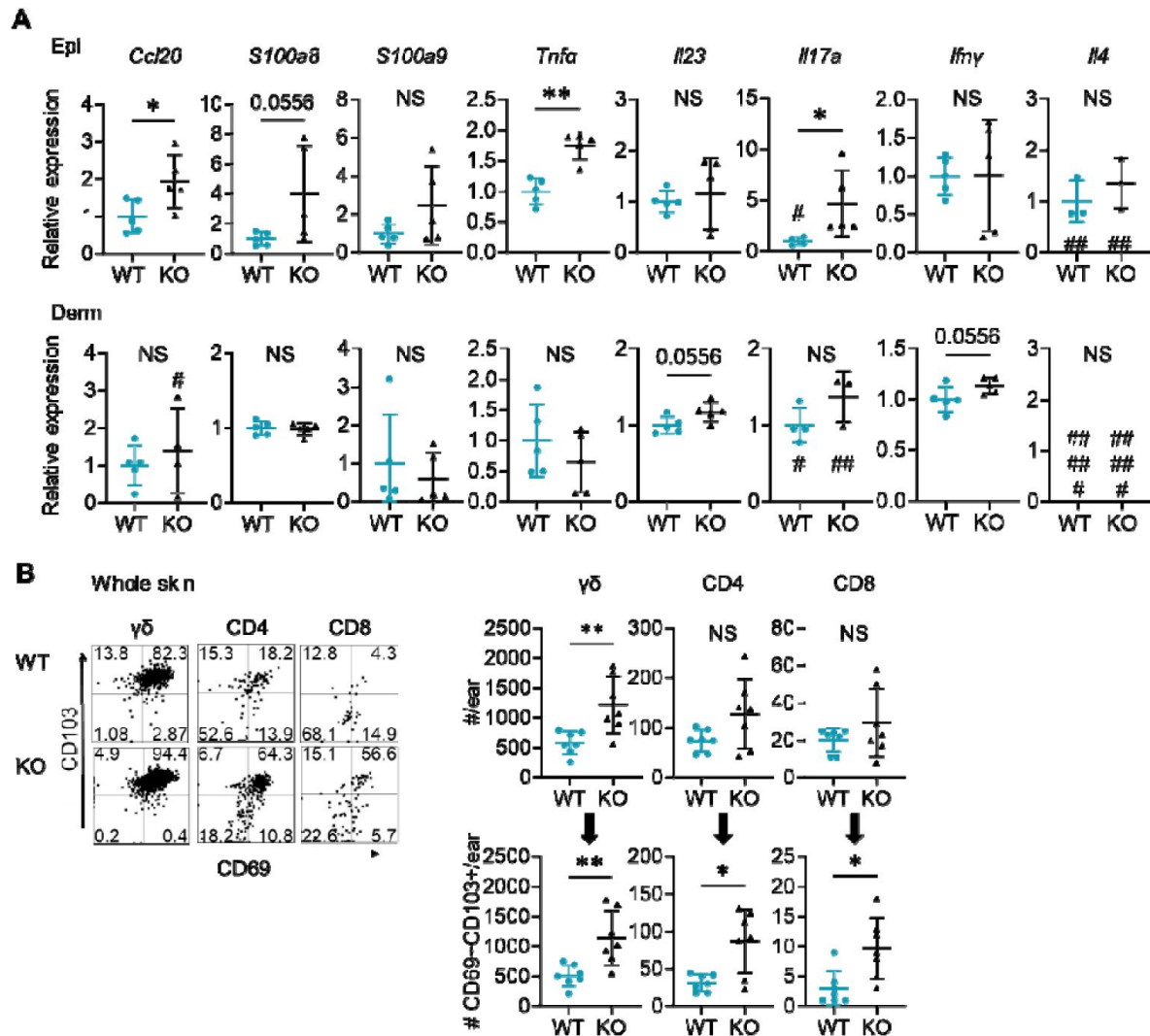


Figure 4-figure supplement 1

Naive Sema4AKO skin shows upregulation of psoriasis related genes and an increase in resident memory T cells.

(A) Relative expression of psoriasis-associated genes in Epi (top) and Derm (bottom) of WT mice and KO mice ($n = 5$ per group, #: not detected). (B) Representative dot plots showing CD69 and CD103 expression in the indicated T cell fractions from whole skin. The graphs show T cell counts per ear (top) and those with resident memory phenotype (bottom) ($n = 7$ per group). Each dot represents the average of 4 ear specimens. A-B: * $p < 0.05$, ** $p < 0.01$. NS, not significant. Figure 4-figure supplement 1-source data 1 Excel file containing quantitative data for Figure 1-figure supplement 1.

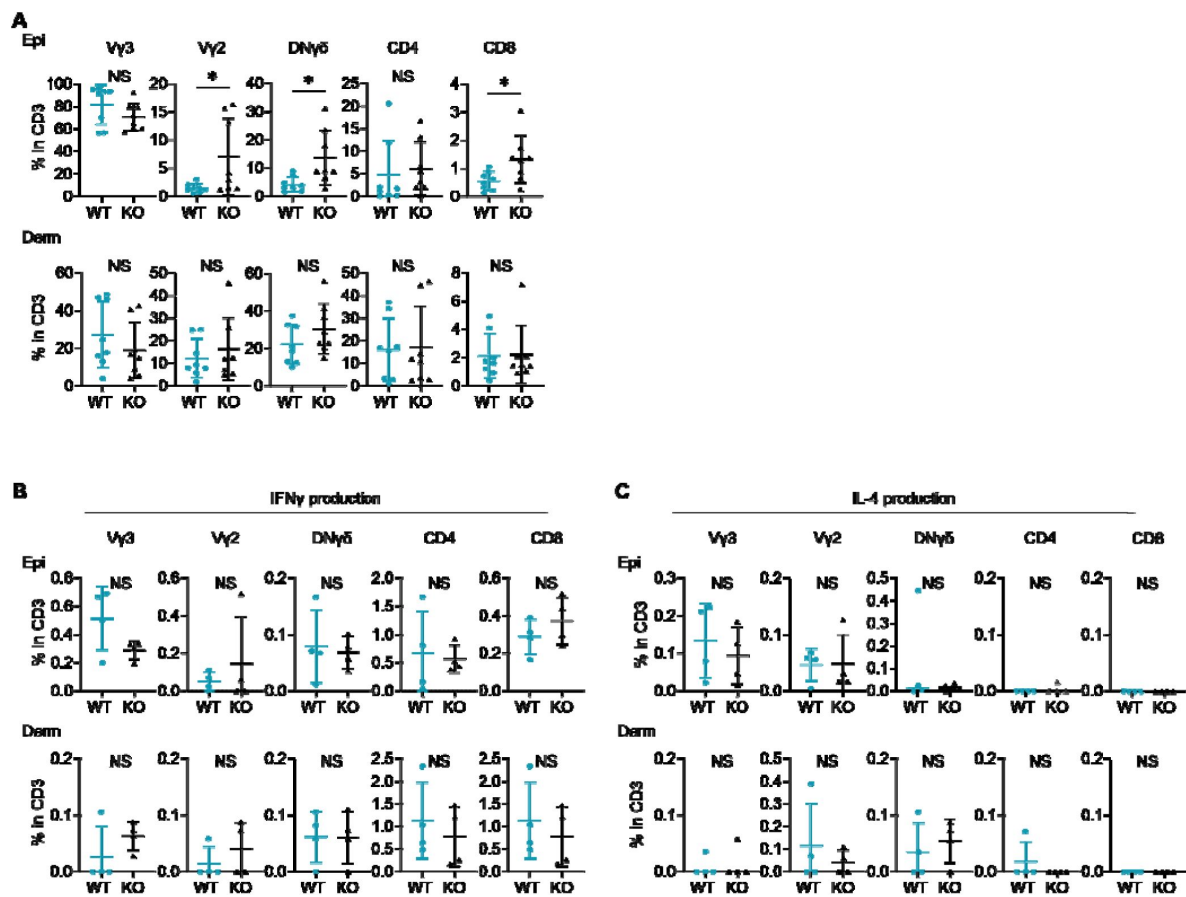


Figure 4-figure supplement 2

Expression of IFNγ and IL-4 is comparable between naive WT and Sema4AKO skin.

(A) The percentages of Vγ3, Vγ2, DNγδ, CD4, and CD8 T cells in CD3 fraction from naive WT and KO mice. (B and C) The graphs presenting the percentages of IFNγ (B) and IL-4 (C) -producing Vγ2, DNγδ, CD4, and CD8 T cells in CD3 fraction in the Epi (top) and Derm (bottom) of naive WT mice and KO mice ($n = 4$ per group). A-C: Each dot represents the average of 4 ear specimens. * $p < 0.05$. NS, not significant. Figure 4-figure supplement 2-source data 1 Excel file containing quantitative data for Figure 4-figure supplement 2.

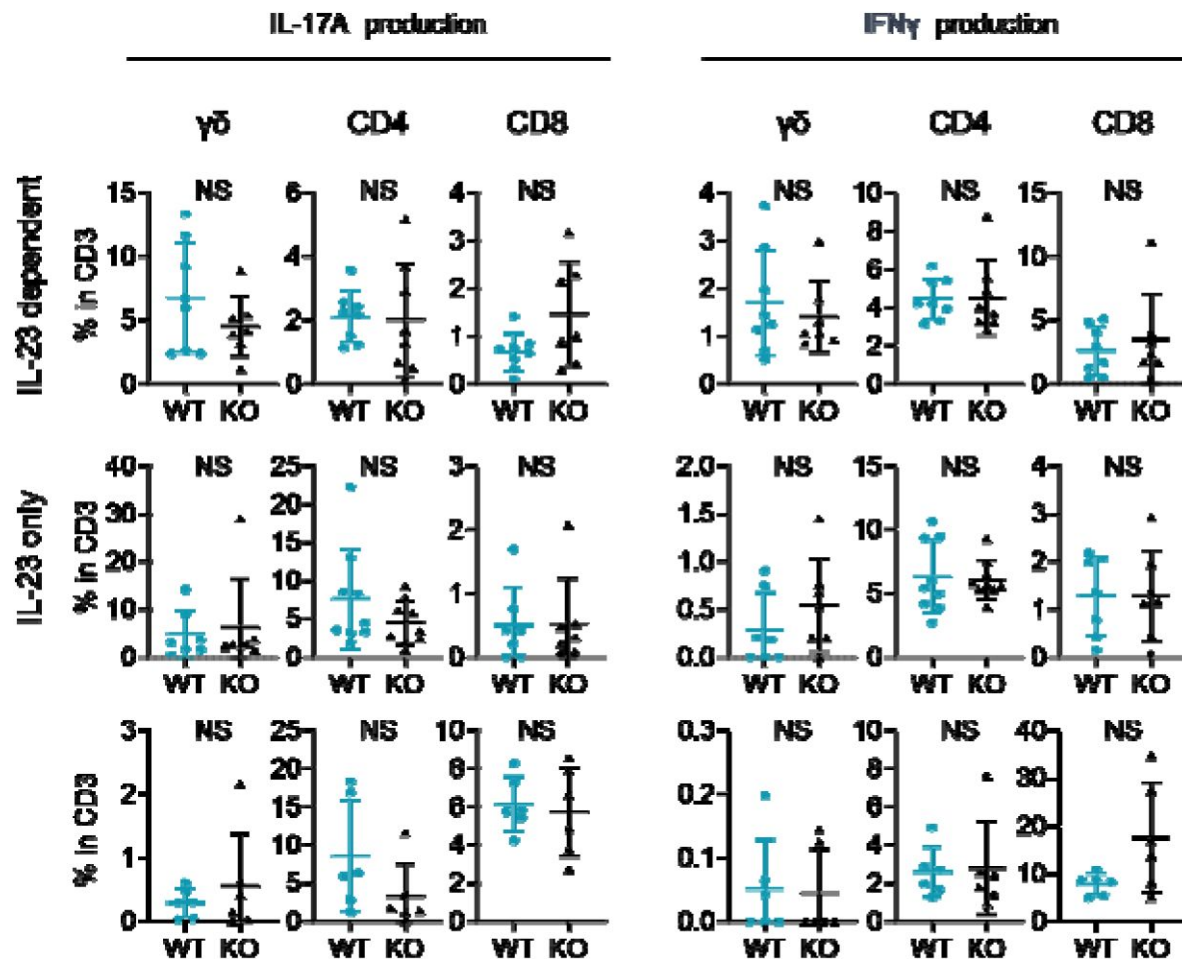


Figure 4-figure supplement 3

Comparable T17differentiation potential under Th17-skewing conditions between WT mice and Sema4AKO mice.

Splenic T cells were cultured for 2 weeks, followed by flow cytometry analysis. The accumulated data display the percentages of IL-17A-producing (right) and IFNγ- producing (left) γδ, CD4, and CD8 T cells within CD3 fraction under various conditions: IL-23 dependent Th17-skewing condition (top), IL-23 only Th17-skewing condition (middle), and IL-23 independent Th17-skewing condition (bottom) . NS, not significant. Figure 4-figure supplement 3-source data 1 Excel file containing quantitative data for Figure 4-figure supplement 3.

Figure 5-figure supplement 1

The epidermis of psoriatic non-lesion is thicker than that of control skin.

Epidermal thickness of Ctl and psoriatic NL ($n = 10$ per group). *** $p < 0.001$. Figure 5-figure supplement1-source data1 Excel file containing quantitative data for Figure 5-figure supplement1.

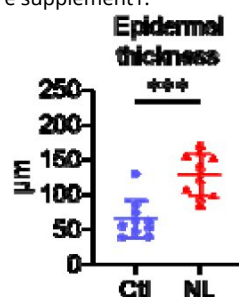
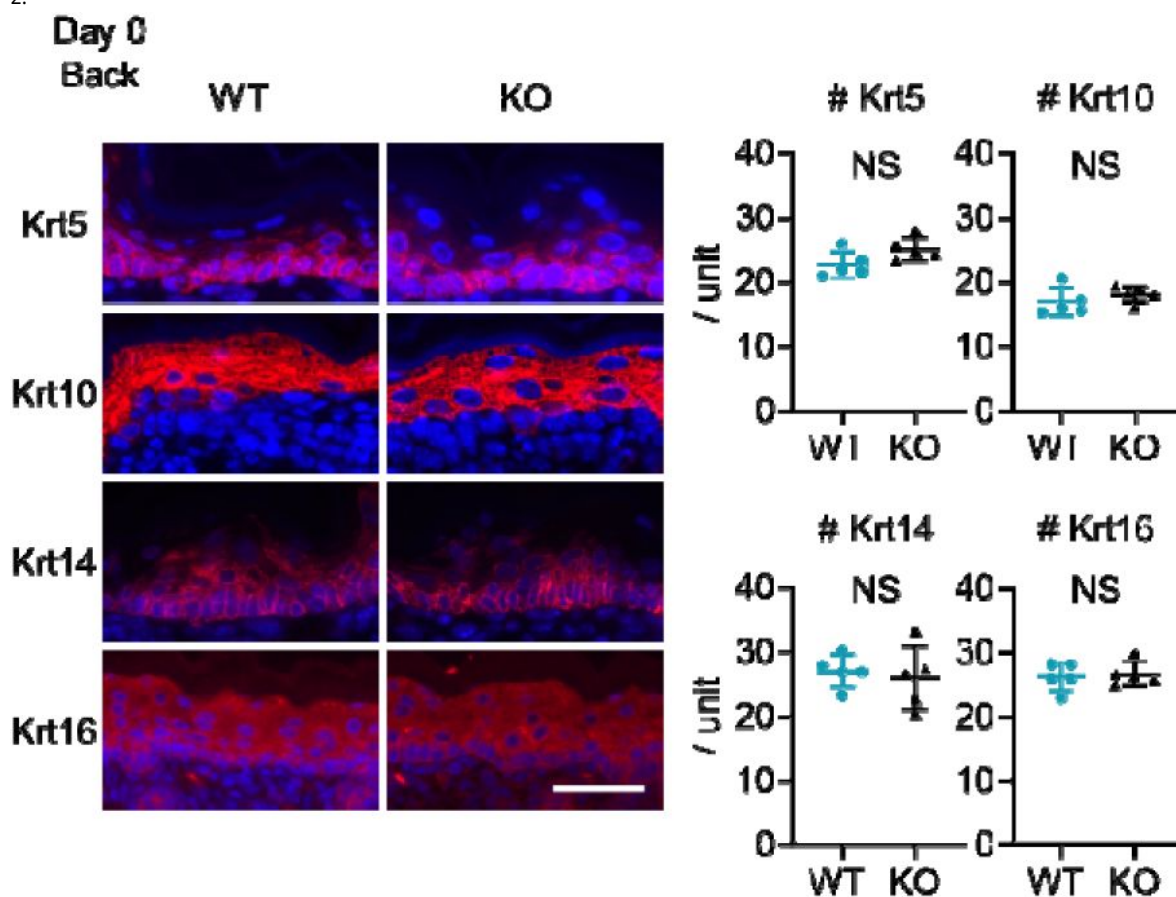


Figure 5-figure supplement 2

Upregulation of cytokeratin expression related to psoriasis is not detected at birth in Sema4AKO mice.

Representative immunofluorescence pictures of Krt5, Krt10, Krt14, and Krt16 (red) overlapped with DAPI, and the accumulated graphs showing the numbers of Krt5, Krt10, Krt14, and Krt16 positive cells per 100 μm width ($n = 5$ per group) in the epidermis of Day 0 back. Scale bar = 50 μm. Each dot represents the average from 5 unit areas per sample. NS, not significant. Figure 5-figure supplement 2-source data 1 Excel file containing quantitative data for Figure 5-figure supplement 2.



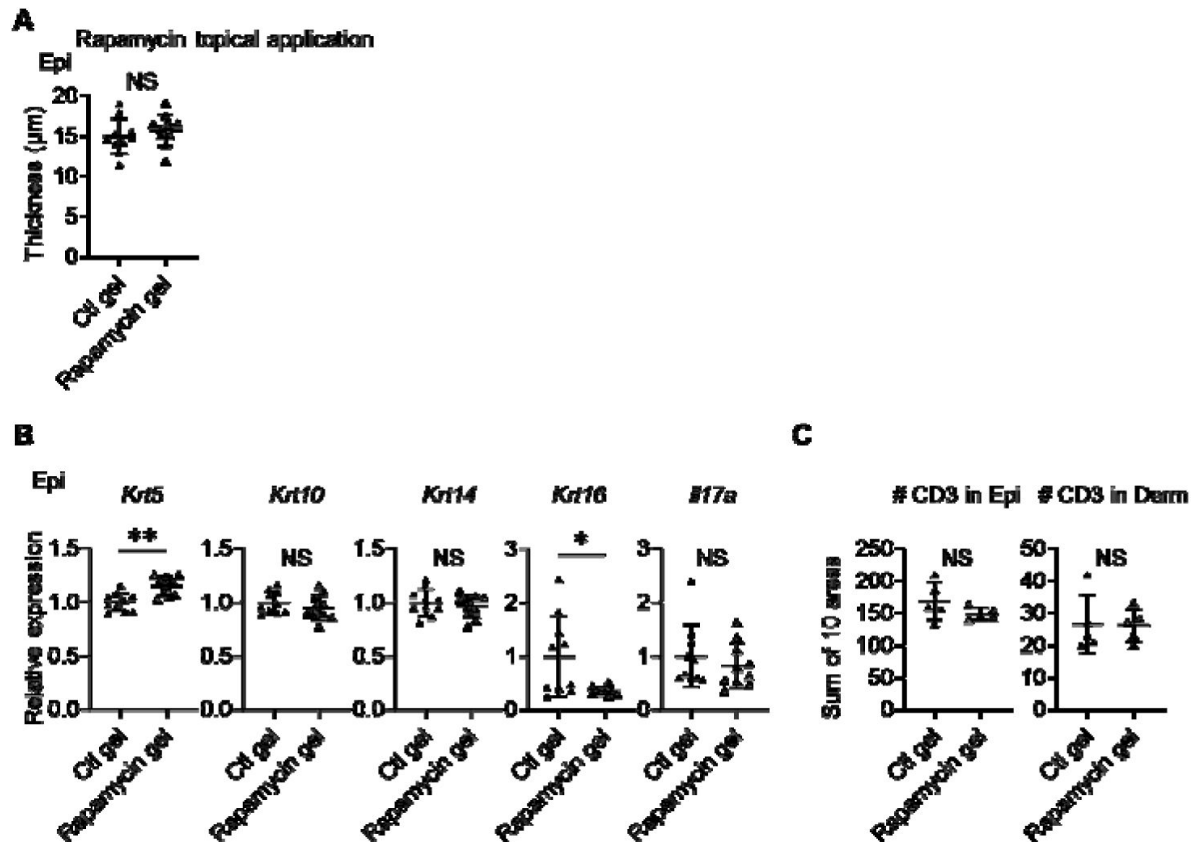


Figure 7-figure supplement 1

Topical application of rapamycin gel yields partially similar results to intraperitoneal treatment.

(A) Comparison of Epi thickness between vehicle (Ctl) gel-treated right ears and rapamycin gel-treated left ears of *Sema4AKO* mice ($n = 10$ per group). (B) Relative expression of keratinocyte differentiation markers and *Il17a* in *Sema4AKO* Epi under Ctl gel or rapamycin gel treatments ($n = 5$ per group). (C) The number of T cells in the Epi (left) and Derm (right), under Ctl gel or Rapamycin gel treatments ($n = 5$ per group). Each dot represents the sum of numbers from 10 unit areas across 3 specimens. A-C: * $p < 0.05$, ** $p < 0.01$. NS, not significant. Figure 7-figure supplement 1-source data 1 Excel file containing quantitative data for Figure 7-figure supplement 1.

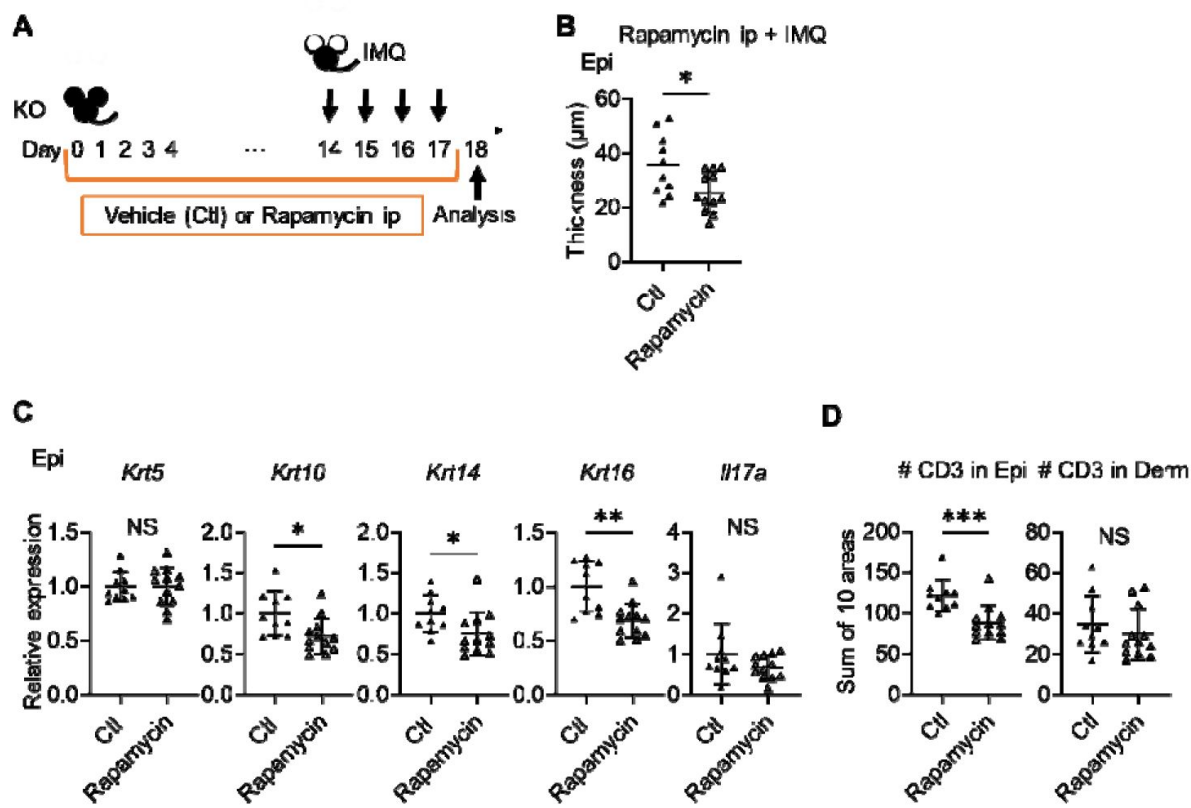


Figure 7-figure supplement 2

Rapamycin treatment reduced the epidermal swelling observed in IMQ-treated Sema4AKO mice.

(A) Experimental scheme. (B) The Epi thickness on Day 18. ($n = 10$ for Ctl, $n = 12$ for Rapamycin). (C) Relative expression of keratinocyte differentiation markers and *Il17a* in Sema4AKO Epi ($n = 10$ for Ctl, $n = 12$ for Rapamycin). (D) The number of T cells in the Epi (left) and Derm (right), under Ctl or rapamycin and IMQ treatments ($n = 10$ for Ctl, $n = 12$ for Rapamycin). Each dot represents the sum of numbers from 10 unit areas across 3 specimens. A-C: * $p < 0.05$, ** $p < 0.01$. NS, not significant. Figure 7-figure supplement 2-source data 1 Excel file containing quantitative data for Figure 7-figure supplement 2.

Fig. 1, D and E Analysis of Sema4A expression by immunohistochemistry

Fig. S4 Epidermal thickness

Psoriasis (n = 10)

Age	Gender	Body site	Severity
66	Male	Leg	Severe
55	Male	Trunk	Moderate
80	Female	Trunk	Severe
72	Male	Trunk	Moderate
30	Male	Trunk	Severe
72	Female	Leg	Moderate
63	Male	Arm	Mild
38	Male	Trunk	Mild
35	Male	Leg	Mild
71	Male	Trunk	Moderate

Control (n = 10)

Age	Gender	Body site
76	Male	Neck
52	Female	Leg
66	Female	Leg
33	Male	Leg
68	Female	Trunk
57	Female	Leg
65	Male	Trunk
41	Female	Trunk
51	Female	Trunk
66	Female	Trunk

Fig 1F Analysis of Sema4A expression by qRT-PCR

Epidermis

Psoriasis (n = 7)

Age	Gender	Body site	Severity
71	Male	Leg	Moderate
76	Female	Neck	Mild
38	Female	Trunk	Moderate
49	Female	Trunk	Moderate
61	Male	Arm	Mild
68	Female	Trunk	Moderate
72	Male	Leg	Mild

Control (n = 10)

Age	Gender	Body site
57	Female	Leg
69	Female	Trunk

Table S1

Patient information

66	Female	Trunk
86	Female	Trunk
53	Female	Trunk
84	Male	Neck
37	Female	Trunk
46	Female	Trunk
51	Female	Trunk
46	Female	Trunk

Dermis

Psoriasis (n = 6)

Age	Gender	Body site	Severity
76	Female	Neck	Mild
38	Female	Trunk	Moderate
49	Female	Trunk	Moderate
61	Male	Arm	Severe
68	Female	Trunk	Moderate
72	Male	Leg	Severe

Control (n = 6)

Age	Gender	Body site
69	Female	Trunk
86	Female	Trunk
84	Male	Neck
37	Female	Trunk
51	Female	Trunk
46	Female	Trunk

Fig 1G Analysis of Sema4A expression by flow cytometry

	Cell type	Average of age	Male (n =)	Female (n =)
Psoriasis (n = 13)	CD4 T cell, CD8 T cell, and monocyte	56.1	9	4
Control (n = 13)	CD4 T cell and CD8 T cell	52.2	6	7
Control (n = 11)	Monocyte	54.1	6	5

Fig 1H Analysis of Serum Sema4A by ELISA

	Average of age	Male (n =)	Female (n =)
Psoriasis (n = 60)	55.3	43	17
Control (n = 20)	52.1	11	9

Fig 6A Pictures of p-S6, S6, p-Akt, and Akt staining

Psoriasis (n = 1)

Age	Gender	Body site	Severity
30	Male	Trunk	Severe

Control (n = 1)

Age	Gender	Body site
66	Female	Trunk

Table S1 (continued)

Analysis of p-S6, S6 expression by immunohistochemistry

Psoriasis (n = 9)

Age	Gender	Body site	Severity
66	Male	Leg	Severe
55	Male	Trunk	Moderate
80	Female	Trunk	Severe
72	Male	Trunk	Moderate
30	Male	Trunk	Severe
72	Female	Leg	Moderate
38	Male	Trunk	Mild
35	Male	Leg	Mild
71	Male	Trunk	Moderate

Control (n = 9)

Age	Gender	Body site
76	Male	Neck
66	Female	Leg
33	Male	Leg
68	Female	Trunk
57	Female	Leg
65	Male	Trunk
41	Female	Trunk
51	Female	Trunk
66	Female	Trunk

Table S1 (continued)

Table S2

Antibodies used for immunohistochemical, immunofluorescence, and western blot analyses

Target	Clone	Species	Supplier	Catalog	Dilution ratio		Incubation time	Incubation temperature
					IHC, IF	WB		
Sema4A	Polyclonal	Rabbit	Abcam	ab70178	100		30 min	RT
phospho-S6 (Ser235/236)	D57.2.2E	Rabbit	Cell Signaling	4858	400	2000	overnight	4 °C
S6	5G10	Rabbit	Cell Signaling	2217	100	1000	overnight	4 °C
phospho-Akt (Ser473)	D9E	Rabbit	Cell Signaling	4060	50	1000	overnight	4 °C
Akt	C67E7	Rabbit	Cell Signaling	4691	300	1000	overnight	4 °C
Keratin 5	Polyclonal	Rabbit	BioLegend	905503	800		overnight	4 °C
Keratin 10	Polyclonal	Rabbit	BioLegend	905403	400		overnight	4 °C
Keratin 14	Polyclonal	Rabbit	BioLegend	905303	400		overnight	4 °C
Cytokeratin 16	8L6R4	Rabbit	Invitrogen	MA5-42892	100		overnight	4 °C
CD3	CD3-12	Rat	Bio-Rad	MCA1477	100		overnight	4 °C
Rabbit IgG H&L (Alexa Fluor 555)	Polyclonal	Donkey	Abcam	ab150074	1000		30 min	RT
Rat IgG H&L (Alexa Fluor 555)	Polyclonal	Donkey	Abcam	ab150154	1000		30 min	RT
β-actin	AC-15	Mouse	Sigma-Aldrich	A5441		5000	overnight	4 °C
Anti-Mouse IgG, HRP-Linked Whole Ab	Monoclonal secondary	Sheep	Cytiva	NA931		10000	1 hour	RT
Anti-Rabbit IgG, HRP-Linked Whole Ab	Polyclonal secondary	Donkey	Cytiva	NA934		10000	1 hour	RT

RT: Room temperature
IHC: immunohistochemistry
IF: immunofluorescence
WB: western blot

Table S3

Primer sequences of real-time quantitative PCR used in human sample experiments

Gene Symbol	Forward primer (5'-3')	Reverse primer (5'-3')
<i>GAPDH</i>	GTCTCCTCTGACTTCAACAGCG	ACCACCCTGTTGCTGTAGCCAA
<i>SEMA4A</i>	TC TGCTCCTGAGTGGTGATG	AAACCAGGACACGGATGAAG

Primer sequences of real-time quantitative PCR in murine sample experiments

Gene Symbol	Forward primer (5'-3')	Reverse primer (5'-3')
<i>Gapdh</i>	TGTGTCCGTCGTGGATCTGA	TTGCTGTGAAGTCGCAGGAG
<i>Krt5</i>	CAGAGCTGAGGAACATGCAG	CATTCTCAGCCGTGGTACG
<i>Krt10</i>	CGTACTGTTTCAGGGTCTGGAG	GCTTCCAGCGATTGTTTCA
<i>Krt14</i>	CGTACTGTTTCAGGGTCTGGAG	GCTTCCAGCGATTGTTTCA
<i>Krt16</i>	TGAGCTGACCCTGTCCAGA	TGAGCTGACCCTGTCCAGA
<i>Filaggrin</i>	GGAGGCATGGTGGAAGTGA	TGTTTATCTTTCCCTCACTTCTACATC
<i>Loricrin</i>	TCACTCATCTTCCCTGGTGCTT	GTCTTTCCACAACCCACAGGA
<i>Sema4a</i>	ATGGCCCTACCATCCCTGG	AGCAGCGTGTCAAAGTCTCG
<i>Ccl20</i>	ATGGCCTGCGGTGGCAAGCGT	CATCTTCTTGACTCTTAGGC
<i>S100a8</i>	TGCGATGGTGATAAAAGTGG	GGCCAGAAGCTCTGCTACTC
<i>S100a9</i>	CACCCTGAGCAAGAAGGAAT	TGTCATTTATGAGGGCTTCATTT
<i>Ifny</i>	ATCTGGAGGAAGTGGCAAAA	TTCAAGACTTCAAAGAGTCTGAGGTA
<i>Tnfa</i>	GCCTCCCTCTCATCAGTTCT	CACTTGGTGGTTTGCTACGA
<i>Il4</i>	ACAGGAGAAGGGACGCCAT	GAAGCCCTACAGACGAGCTCA
<i>Il17a</i>	CTGTGTCTCTGATGCTGTTG	ATGTGGTGGTCCAGCTTTC
<i>Il23</i>	TCCCTACTAGGACTCAGCCAAC	GCTGCCACTGCTGACTAGAA

Table S4**Antibodies used for flow cytometry analysis**

Target	Clone	Species	Supplier	Catalog
CD3	OKT3	Mouse	BioLegend	317335
CD4	RPA-T4	Mouse	BioLegend	300512
CD8a	RPA-T8	Mouse	eBioscience	47-0088-42
CD11c	3.9	Mouse	BioLegend	301628
SEMA4A	5E3/SEMA4A	Mouse	BioLegend	148404
CD3ε	145-2C11	Armenian Hamster	BioLegend	100328
CD4	GK1.5	Rat	BioLegend	100406
CD8a	53-6.7	Rat	BioLegend	100714
CD16/32	93	Rat	BioLegend	101301
CD69	H1.2F3	Armenian Hamster	BioLegend	104514
CD103	2E7	Armenian Hamster	BioLegend	121422
TCR Vγ2	UC3-10A6	Armenian Hamster	BioLegend	137705
TCR Vγ3	536	Syrian Hamster	BD Biosciences	743241
TCRγδ	GL3	Armenian Hamster	BioLegend	118124
IFNγ	XMG1.2	Rat	BioLegend	505813
IL-4	11B11	Rat	BD Biosciences	562915
IL-17A	TC11-18H10.1	Rat	BioLegend	506925

Table S5**Mouse recombinant cytokines**

Cytokines	Supplier	Catalog
IL-1β	BioLegend	575102
IL-6	BioLegend	575702
IL-23	BioLegend	589002
TGFβ1	BioLegend	763102

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Reviewer #1 (Public review):

Summary:

In this study, Kume et al examined the role of the protein Semaphorin 4a in steady state skin homeostasis and how this relates to skin changes seen in human psoriasis and imiquimod-induced psoriasis-like disease in mice. The authors found that human psoriatic skin has reduced expression of Sema4a in the epidermis. While Sema4a has been shown to drive inflammatory activation in different immune populations, this finding suggested Sema4a might be important for negatively regulating Th17 inflammation in the skin. The authors go on to show that Sema4a knockout mice have skin changes in key keratinocyte genes, increased gdT cells, and increased IL-17 similar to differences seen in non-lesional psoriatic skin, and that bone marrow chimera mice with WT immune cells and Sema4a KO stromal cells develop worse IMQ-induced psoriasis-like disease, further linking expression of Sema4a in the skin to maintaining skin homeostasis. The authors next studied downstream pathways that might mediate the homeostatic effects of Sema4a, focusing on mTOR given its known role in keratinocyte function. Like for the immune phenotypes, Sema4a KO mice had increased mTOR activation in the epidermis in a similar pattern to mTOR activation noted in non-lesional psoriatic skin. The authors next targeted the mTOR pathway and showed rapamycin could reverse some of the psoriasis-like skin changes in Sema4a KO mice, confirming the role of increased mTOR in contributing to the observed skin phenotype.

In the revised manuscript, the authors expand on the potential relevance to psoriasis by demonstrating similar findings in an IL-23-driven model of skin inflammation, which is an orthogonal model of psoriasis to their original IMQ model. They also show that in addition to reversing steady state differences in skin thickness between Sema4a KO mice and WT mice, rapamycin improves metrics of disease in the IMQ model of psoriasis. These additional studies further bolster their conclusions that Sema4a may play a protective role in by preventing over-activation of mTOR in the skin in psoriasis.

Strengths:

The most interesting finding is the tissue-specific role for Sema4a, where it has previously been considered to play a mostly pro-inflammatory role in immune cells, this study shows that when expressed by keratinocytes, Sema4a plays a homeostatic role that when missing leads to development of psoriasis-like skin changes. This has important implications in terms of targeting Sema4a pharmacologically. It also may yield a novel mouse model to study mechanisms of psoriasis development in mice separate from the commonly used IMQ model. The included experiments are well-controlled and executed rigorously.

The new experiments provide additional data to support the conclusions through an orthogonal model of psoriasis and demonstrating rapamycin-induced reversal of changes in the IMQ disease model.

Weaknesses:

While the main weakness of these studies, lack of tissue-specific Sema4a knockout mice (e.g. in keratinocytes only), remains, generating these mice and performing the necessary experiments is beyond the scope of completing these particular studies. Similarly, it is understandable that additional bone marrow chimeras would be costly and labor intensive without adding much more in the absence of tissue-specific knockouts.

<https://doi.org/10.7554/eLife.97654.2.sa2>

Reviewer #2 (Public review):

Summary:

Kume et al. found for the first time that Semaphorin 4A (Sema4A) was downregulated in both mRNA and protein levels in L and NL keratinocytes of psoriasis patients compared to control keratinocytes. In peripheral blood, they found that Sema4A is not only expressed in keratinocytes but is also upregulated in hematopoietic cells such as lymphocytes and monocytes in the blood of psoriasis patients. They investigated how the down-regulation of Sema4A expression in psoriatic epidermal cells affects the immunological inflammation of psoriasis by using a psoriasis mice model in which Sema4A KO mice were treated with IMQ. Kume et al. hypothesized that down-regulation of Sema4A expression in keratinocytes might be responsible for the augmentation of psoriasis inflammation. Using bone marrow chimeric mice, Kume et al. showed that KO of Sema4A in non-hematopoietic cells was responsible for the enhanced inflammation in psoriasis. The expression of CCL20, TNF, IL-17, and mTOR was upregulated in the Sema4AKO epidermis compared to the WT epidermis, and the infiltration of IL-17-producing T cells was also enhanced.

Strengths:

Decreased Sema4A expression may be involved in psoriasis exacerbation through epidermal proliferation and enhanced infiltration of Th17 cells, which helps understand psoriasis immunopathogenesis.

Weaknesses:

The mechanism of decreased Sema4A expression in psoriasis is not clear, although this does not affect the strength of this research.

<https://doi.org/10.7554/eLife.97654.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public reviews:

Reviewer #1 (Public Review):

Summary:

In this study, Kume et al examined the role of the protein Semaphorin 4a in steady-state skin homeostasis and how this relates to skin changes seen in human psoriasis and imiquimod-induced psoriasis-like disease in mice. The authors found that human psoriatic skin has reduced expression of Sema4a in the epidermis. While Sema4a has been shown to drive inflammatory activation in different immune populations, this finding suggested Sema4a might be important for negatively regulating Th17

inflammation in the skin. The authors go on to show that Sema4a knockout mice have skin changes in key keratinocyte genes, increased gdT cells, and increased IL-17 similar to differences seen in non-lesional psoriatic skin, and that bone marrow chimera mice with WT immune cells and Sema4a KO stromal cells develop worse IMQ-induced psoriasis-like disease, further linking expression of Sema4a in the skin to maintaining skin homeostasis. The authors next studied downstream pathways that might mediate the homeostatic effects of Sema4a, focusing on mTOR given its known role in keratinocyte function. As with the immune phenotypes, Sema4a KO mice had increased mTOR activation in the epidermis in a similar pattern to mTOR activation noted in non-lesional psoriatic skin. The authors next targeted the mTOR pathway and showed rapamycin could reverse some of the psoriasis-like skin changes in Sema4a KO mice, confirming the role of increased mTOR in contributing to the observed skin phenotype.

Strengths:

The most interesting finding is the tissue-specific role for Sema4a, where it has previously been considered to play a mostly pro-inflammatory role in immune cells, this study shows that when expressed by keratinocytes, Sema4a plays a homeostatic role that when missing leads to the development of psoriasis-like skin changes. This has important implications in terms of targeting Sema4a pharmacologically. It also may yield a novel mouse model to study mechanisms of psoriasis development in mice separate from the commonly used IMQ model. The included experiments are well-controlled and executed rigorously.

Weaknesses:

A weakness of the study is the lack of tissue-specific Sema4a knockout mice (e.g. in keratinocytes only). The authors did use bone marrow chimeras, but only in one experiment. This work implies that psoriasis may represent a Sema4a-deficient state in the epidermal cells, while the same might not be true for immune cells. Indeed, in their analysis of non-lesional psoriasis skin, Sema4a was not significantly decreased compared to control skin, possibly due to compensatory increased Sema4a from other cell types. Unbiased RNA-seq of Sema4a KO mouse skin for comparison to non-lesional skin might identify other similarities besides mTOR signaling. Indeed, targeting mTOR with rapamycin reverses some of the skin changes in Sema4a KO mice, but not skin thickness, so other pathways impacted by Sema4a may be better targets if they could be identified. Utilizing WT → KO chimeras in addition to global KO mice in the experiments in Figures 6-8 would more strongly implicate the separate role of Sema4a in skin vs immune cell populations and might more closely mimic non-lesional psoriasis skin.

We sincerely appreciate your summary and for pointing out the strengths and weaknesses of our study. Although we were unfortunately unable to perform all these experiments due to limitations in our resources, we fully agree with the importance of studying tissue-specific Sema4A KO mice. As an alternative, we compared the IL-17A-producing potential of skin T cells between WT → KO mice and KO → KO mice following 4 consecutive days of IMQ treatment using flow cytometry. The results were comparable between the two groups. Additionally, we performed RNA-seq on the epidermis of WT and Sema4A KO mice. While we did not find similarities between Sema4A KO skin and non-lesional psoriasis except for *S100a8* expression, we will further try to seek for the mechanisms how Sema4A KO skin mimics non-lesional psoriasis skin as a future project.

Although targeting mTOR with rapamycin did not reverse the epidermal thickness in Sema4A KO mice, rapamycin was effective in reducing epidermal thickness in a murine psoriasis model induced by IMQ in Sema4A KO mice. These results suggest potential clinical relevance for treating active, lesional psoriatic skin changes, which would be of interest to clinicians. Thank you once again for your valuable insights.

Reviewer #2 (Public Review):

Summary:

Kume et al. found for the first time that Semaphorin 4A (Sema4A) was downregulated in both mRNA and protein levels in L and NL keratinocytes of psoriasis patients compared to control keratinocytes. In peripheral blood, they found that Sema4A is not only expressed in keratinocytes but is also upregulated in hematopoietic cells such as lymphocytes and monocytes in the blood of psoriasis patients. They investigated how the down-regulation of Sema4A expression in psoriatic epidermal cells affects the immunological inflammation of psoriasis by using a psoriasis mice model in which Sema4A KO mice were treated with IMQ. Kume et al. hypothesized that down-regulation of Sema4A expression in keratinocytes might be responsible for the augmentation of psoriasis inflammation. Using bone marrow chimeric mice, Kume et al. showed that KO of Sema4A in non-hematopoietic cells was responsible for the enhanced inflammation in psoriasis. The expression of CCL20, TNF, IL-17, and mTOR was upregulated in the Sema4AKO epidermis compared to the WT epidermis, and the infiltration of IL-17-producing T cells was also enhanced.

Strengths:

Decreased Sema4A expression may be involved in psoriasis exacerbation through epidermal proliferation and enhanced infiltration of Th17 cells, which helps understand psoriasis immunopathogenesis.

Weaknesses:

The mechanism by which decreased Sema4A expression may exacerbate psoriasis is unclear as yet.

We greatly appreciate your summary and thoughtful feedback on the strengths and weaknesses of our study. In response, we have included the results of additional experiments on IL-23-mediated psoriasis-like dermatitis, which showed that epidermal thickness was significantly greater in KO mice compared to WT mice. When we analyzed the T cells infiltrating the ears using flow cytometry, the proportion of IL-17A producing V γ 2 and DN γ 8 T cells within the CD3 fraction of the epidermis was significantly higher in Sema4A KO mice, consistent with the results from IMQ-induced psoriasis-like dermatitis. Furthermore, we examined STAT3 expression in the epidermis of WT and Sema4A KO mice using Western blot analysis, and the results were comparable between the two groups. However, the mechanism by which decreased Sema4A expression may exacerbate psoriasis remains unclear. We have added some explanations and presumptions to the limitations section. Thank you once again for your valuable insights.

Recommendations For The Authors:

Reviewer #1 (Recommendations For The Authors):

Figure 1C

What statistics were used? The supplemental notes adjusted the P value, what correction for multiple comparisons was utilized? Could the authors instead show logFC for the DEGs between Ctl and L in each cluster? This might be best demonstrated with a volcano plot, highlighting SEMA4A, and other genes known to be DE in psoriasis.

We apologize for not including the detailed analysis methods in the original manuscript submission. We analyzed the scRNA-seq data using Cellxgene VIP with Welch's t-test. Multiple

comparisons were performed using the Benjamini-Hochberg procedure, setting the false discovery rate (FDR) at 0.05. These details are now explained in the MATERIALS AND METHODS section of the resubmitted manuscript. We also added a log₂FC-log₁₀ *p*-value graph for the DEGs in keratinocytes between Ctl and L to Figure 1-figure supplement 1D. The log₂FC values in keratinocytes, dendritic cells, and macrophages were -0.07, 0.00, and -0.05, respectively. Although the log₂FC is low in keratinocytes, the adjusted *p*-value (*padj*) for *Sema4A* is 2.83×10⁻³⁹, indicating a statistically significant difference.

Page 8 Line 111 in the resubmitted manuscript:

“The adjusted *p*-value (*padj*) for *SEMA4A* in keratinocytes between Ctl and L was 2.83×10⁻³⁹, indicating a statistically significant difference despite not being visually prominent in the volcano plot, which shows comprehensive differential gene expression in keratinocytes (Figure 1C; Figure 1-figure supplement 1D).”

Page 54: In the Figure legend of Figure 1-figure supplement 1D in the resubmitted manuscript:

“(D) The volcano plot displays changes in gene expression in psoriatic L compared to Ctl.”

Page 30 Line 481 in the resubmitted manuscript: In the “Data processing of single-cell RNA-sequencing and bulk RNA-sequencing” section.

“The data was integrated into an h5ad file, which can be visualized in Cellxgene VIP (K. Li et al., 2022). We then performed differential analysis between two groups of cells to identify differential expressed genes using Welch’s *t*-test. Multiple comparisons were controlled using the Benjamini-Hochberg procedure, with the false discovery rate set at 0.05 and significance defined as *padj* < 0.05.”

Figure 2B

The results narrative notes WT->WT is comparable to KO->WT. No statistics are given for this comparison. It appears the difference is less than the other comparisons, but still may be significant. Also, in the supplemental for Figure 2B, there appear to be missing columns for the 4 BM chimera groups (columns for WT and KO, but not 4 columns for each donor: recipient pair).

We sincerely apologize for any confusion. We presented the results of the chimeric mice in Figure 3, and Figure 3-source data 1 shows the 4 BM chimera groups. In Figure 3B, the *p*-value for the comparison between WT->WT mice and KO->WT mice was 0.7988, as indicated in Figure 3-source data 1.

Figure 3B

While ear skin is not easily obtainable at day 0 for comparison, why not also include back skin at Wk 8? If the back skin epidermis is thicker like the ear skin, it supports the ear skin conclusion and adds a more consistent comparison. If the back skin epidermis is not thicker, what would be the author's explanation as to the why only ear skin epidermis is thicker in KO mice at 8 weeks?

We appreciate and completely agree with the reviewer’s insightful comment. We have added images and dot plots of the back skin at Week 8 in Figure 4B. Since the back skin epidermis is thicker, similar to the ear skin, these results support the conclusion drawn from the ear skin data. Regarding Figure 4C, which shows the expression of *Sema4a* in the epidermis and dermis of 8-week-old WT mouse ear, we have modified the sentence in the manuscript to ‘the epidermis of WT ear at Week 8’ for clarification.

Page 12 Line 180 in the resubmitted manuscript:

“While epidermal thickness of back skin was comparable at birth (Figure 4B), on week 8, epidermis of *Sema4AKO* back and ear skin was notably thicker than that of WT mice (Figure 4B), suggesting that acanthosis in *Sema4AKO* mice is accentuated post-birth.”

Page 47: In the Figure legend of Figure 4B in the resubmitted manuscript:

“(B) Left: representative Hematoxylin and eosin staining of Day 0 back and Wk 8 back and ear. Scale bar = 50 μ m. Right: Epi and Derm thickness in Day 0 back ($n = 5$) and Wk 8 back ($n = 5$) and ear ($n = 8$).”

Figures 3C&D, Figures 4 D-F

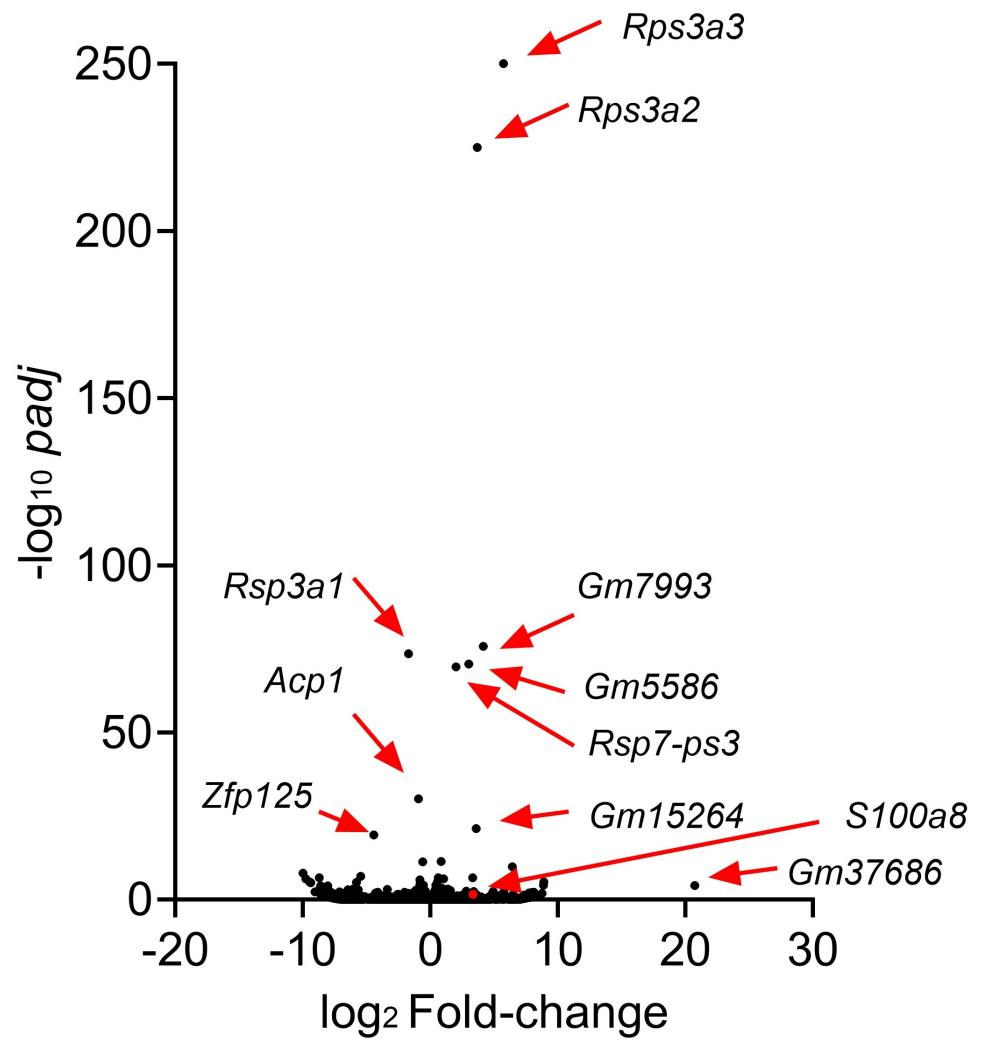
*The figures might be easier to read if some of the data is moved to supplemental, especially in Figure 4, which has 36 panels just in D-F. Conversely, the dLN data is important in establishing the skin microenvironment as important in the accumulation of $\gamma\delta$ cells and IL-17 production in the setting of *Sema4a* KO, so this might be more impactful if moved to the main figure.*

We appreciate and agree with your comments. As recommended, we have moved data from Figure 3C and 4D-F to the supplemental section. The dLN data have been moved to the main figure as Figure 4E. This has improved the readability of the figures.

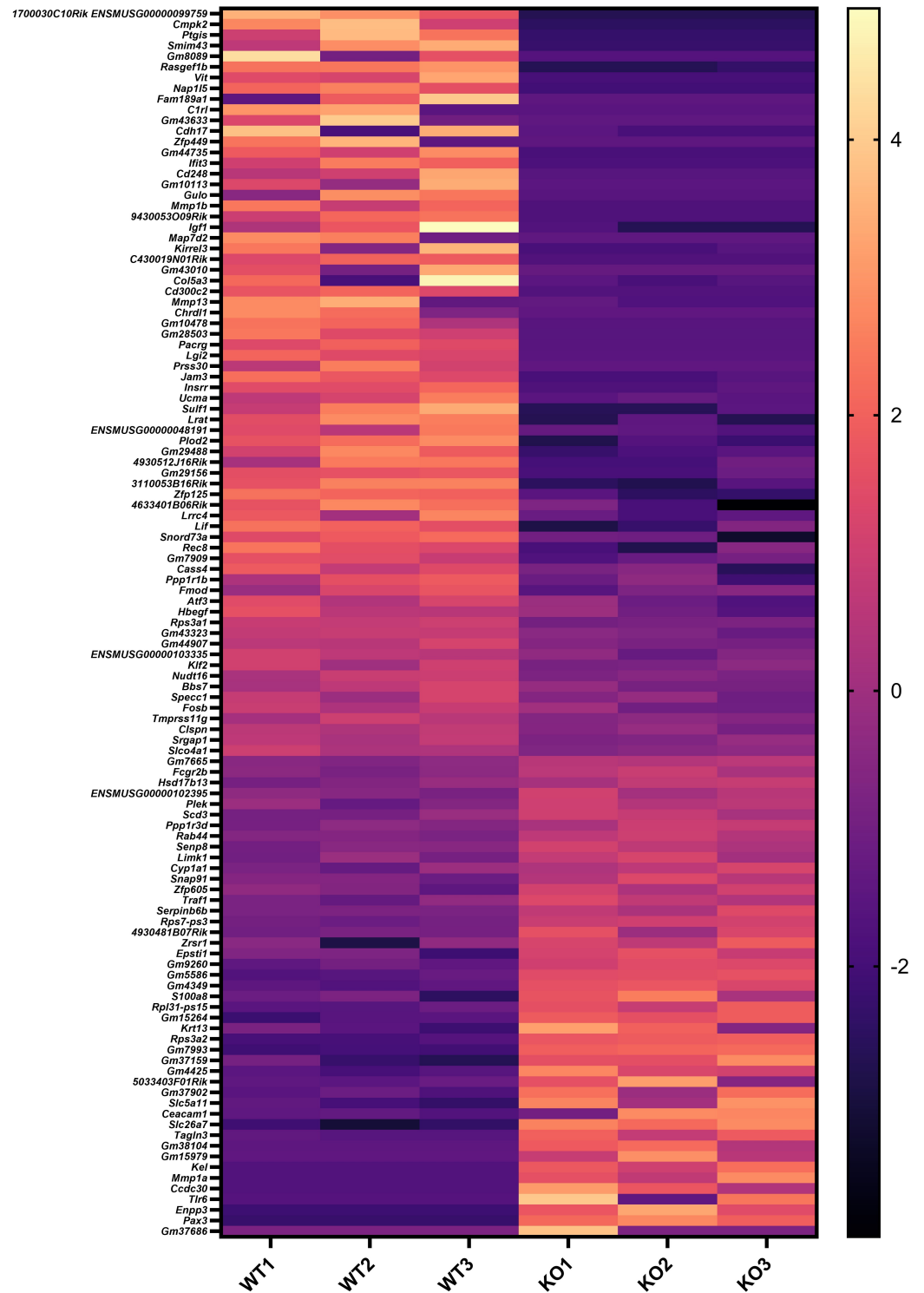
*Figure 5 and Figure 6 might work better if combined. The differences in keratinocytes in psoriasis are well-known, so the novelty is how *Sema4a* KO skin appears to share similar differences. This would be easier to see if compared side-by-side in the same figure. Also, there is an opportunity to show this more rigorously by performing RNA-seq on WT vs *Sema4a* KO skin. Showing a larger set of DEGs that trend similarly between Ctl/NL psoriasis and WT/*Sema4a* KO skin in a heatmap would bolster the conclusion that *Sema4a* deficiency contributes to a psoriasis-like skin defect.*

We appreciate your valuable suggestion. Following your recommendation, we have combined Figures 5 and 6 to facilitate a side-by-side comparison. This highlights the similarities between *Sema4AKO* skin and psoriasis, making it easier to observe differences in keratinocytes. Additionally, we performed RNA-seq on WT and *Sema4a* KO epidermis ($n = 3$ per group). We analyzed the raw count data using iDEP 2.0 (Ge S.X., *BMC Bioinformatics*, 2018), setting the minimal counts per million to 0.5 in at least one library. Differential gene expression analysis was conducted using DESeq2, with an FDR cutoff of 0.1 and a minimum fold change of 2. As a result, we identified 46 upregulated and 70 downregulated genes in *Sema4AKO* mice compared to WT mice (see the volcano plot and heat map). However, except for *S100a8*, we did not observe significant expression changes in non-lesional psoriasis-related genes between WT and *Sema4AKO* mice. In the future, we aim to identify subtle stimuli that could cause gene expression changes between these groups and we would like to perform additional RNA-seq experiments.

Author response image 1.



Author response image 2.



Page 48: The Figure title of Figure 5 in the resubmitted manuscript:

“Figure 5: Sema4AKO skin shares the features of human psoriatic NL.”

SEMA4A is not significantly DE between Ctl and NL in the psoriasis RNA-seq data. If a lower expression of SEMA4A in psoriasis skin is a driving part of the phenotype, why is this not observed in the RNA-seq data? Presumably, this could be explained by infiltration of immune cells with increased SEMA4A expression, like in the scRNA-seq data in Figure 1. If so, might it be useful to analyze WT→KO chimera mice similarly to global KO mice in Figures 6-8? This might more accurately reflect what is happening in psoriasis, if epidermal SEMA4A expression is low, but immune expression is not. The KO data on their own nicely show a skin phenotype, but these additional experiments might more closely mimic psoriatic disease and increase the rigor and impact of the study.

We really appreciate your insightful comments. Due to the limitations of the animal experimentation facility, we regret that we are unable to create additional chimeric mice. Although our analysis is limited, we compared IL-17A production from T cells of WT → KO mice and KO → KO mice following 4 consecutive days of IMQ treatment using flow cytometry (see Author response image 3 below; $n = 6$ for WT → KO, $n = 4$ for KO → KO). This comparison revealed that IL-17A production from T cells was comparable, regardless of whether they were derived from WT or Sema4AKO mice, when the skin constituent cells were derived from Sema4AKO. We appreciate the value of your advice, and agree that investigating keratinocyte differentiation and mTOR signaling in the epidermis, using either WT → KO chimeric mice or keratinocyte-specific Sema4A-deficient mice, is a crucial next step in our research.

Author response image 3.

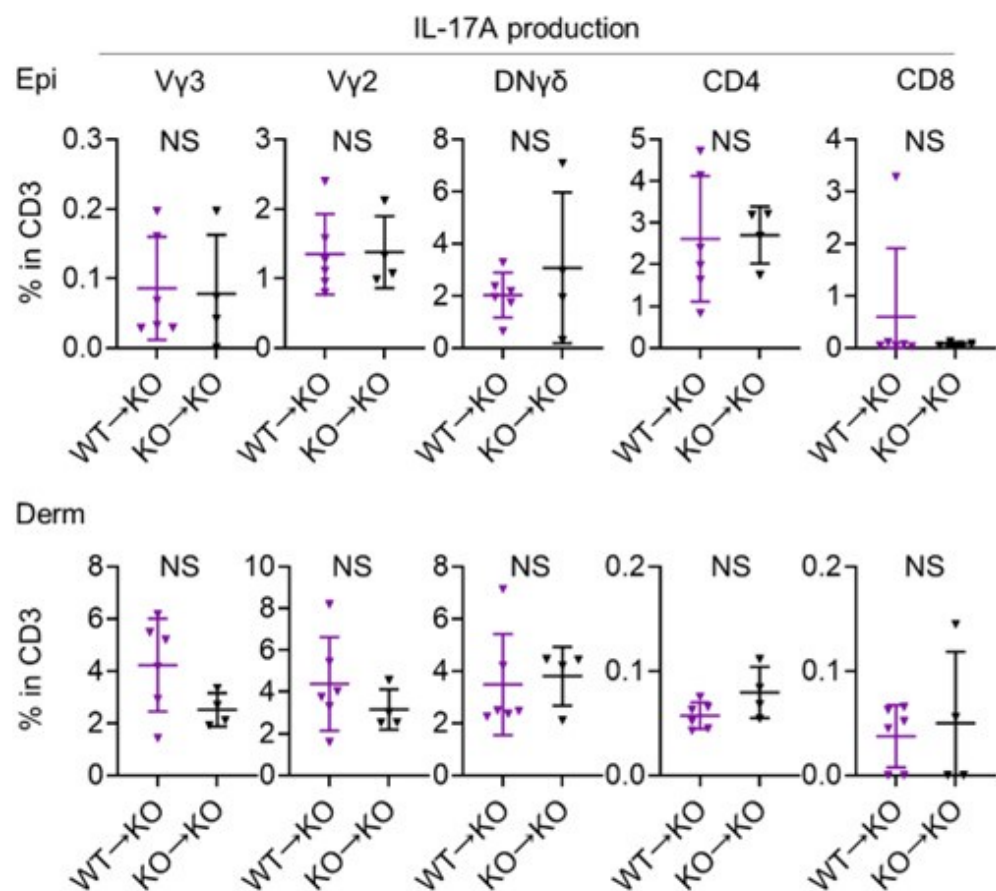


Figure 8

Rapamycin was able to partially reverse the psoriasis-like skin phenotype in Sema4a KO mice. Would rapamycin also be effective in the more severe disease induced by IMQ in Sema4a KO mice? While partially reducing the effect of Sema4a KO on steady-state skin with rapamycin strengthens the link to mTOR dysregulation, it did not change skin thickness. It's unclear if this would be useful clinically for patients with well-controlled psoriasis (NL skin). Would it be useful to reverse active, lesional psoriatic skin changes? Testing this might yield results more relevant to clinicians and patients.

We are grateful for your valuable feedback. Rapamycin showed effectiveness in reducing epidermal thickness in a murine psoriasis model induced by IMQ in Sema4AKO mice. Rapamycin treatment downregulated the expression of *Krt10*, *Krt14*, and *Krt16*. We included these results to Figure 7-figure supplement 2. These results suggest potential clinical relevance for treating active, lesional psoriatic skin changes and may be of interest to clinicians and patients.

Page 17 Line 269 in the resubmitted manuscript:

“Next, we investigated whether intraperitoneal rapamycin treatment effectively downregulates inflammation in the IMQ-induced murine model of psoriasis in Sema4AKO mice (Figure 7-figure supplement 2A). Rapamycin significantly reduced epidermal thickness compared to vehicle treatment (Figure 7-figure supplement 2B). Additionally, rapamycin treatment downregulated the expression of *Krt10*, *Krt14*, and *Krt16* (Figure 7-figure supplement 2C). While the upregulation of *Il17a* in the Sema4AKO epidermis in IMQ model was not clearly modified by rapamycin (Figure 7-figure supplement 2C), immunofluorescence revealed a decrease in the number of CD3 T cells in Sema4AKO epidermis by rapamycin (Figure 7-figure supplement 2D). In the naive states, mTORC1 primarily regulates keratinocyte proliferation, whereas mTORC2 mainly involved in the keratinocyte differentiation through Sema4A-related signaling pathways. Conversely, in the psoriatic dermatitis state, rapamycin downregulated both keratinocyte differentiation and proliferation markers. The observed similarities in *Il17a* expression following treatment with rapamycin and JR-AB2-011, regardless of additional IMQ treatment, suggest that *Il17a* production is not significantly dependent on Sema4A-related mTOR signaling.”

Page 29 Line 461 in the resubmitted manuscript: In the “Inhibition of mTOR” section.

“To analyze the preventive effectiveness of rapamycin in an IMQ-induced murine model of psoriatic dermatitis, Sema4AKO mice were administered either vehicle or rapamycin intraperitoneally from Day 0 to Day 17, and IMQ was topically applied to both ears for 4 days starting on Day 14. Then, on Day 18, ears were collected for further analysis.”

Page 71: Figure 7-figure supplement 2 in the resubmitted manuscript:

“Figure 7-figure supplement 2: Rapamycin treatment reduced the epidermal swelling observed in IMQ-treated Sema4AKO mice.

(A) Experimental scheme. (B) The Epi thickness on Day 18. ($n = 10$ for Ctl, $n = 12$ for Rapamycin). (C) Relative expression of keratinocyte differentiation markers and *Il17a* in Sema4AKO Epi ($n = 10$ for Ctl, $n = 12$ for Rapamycin). (D) The number of T cells in the Epi (left) and Derm (right), under Ctl or rapamycin and IMQ treatments ($n = 10$ for Ctl, $n = 12$ for Rapamycin). Each dot represents the sum of numbers from 10 unit areas across 3 specimens. A-C: * $p < 0.05$, ** $p < 0.01$. NS, not significant.”

Reviewer #2 (Recommendations For The Authors):

(1) To know whether the decrease of Sema4A in the epidermis of psoriasis patients is a result or a cause of psoriasis, it is necessary to show how the expression of Sema4A in epidermal cells is regulated. Shouldn't the degree of change in the expression of essential molecules (which is the cause of psoriasis) be more pronounced in L than in NL?

We surveyed transcription factors of human Sema4A using GeneCards and found that NF- κ B is the transcription factor most frequently associated with psoriasis. Wang et al. (*Arthritis Res Ther.* 2015) indicated NF- κ B-dependent modulation of Sema4A expression in synovial fibroblasts of rheumatoid arthritis. However, since NF- κ B expression is reportedly upregulated in psoriasis lesions, other transcription factors may function as key modulators of Sema4A expression in the epidermis.

Although the molecules causing psoriasis remain to be elucidated, we investigated the correlation between the expression of psoriasis-related essential molecules in keratinocytes—such as *S100A7A*, *S100A7*, *S100A8*, *S100A9*, and *S100A12*—and *SEMA4A* expression in L and NL samples using qRT-PCR. We could not identify a correlation between these molecules and *SEMA4A* expression. We added a note to the limitations section to acknowledge that we were not able to reveal how Sema4A expression is regulated and that we could not determine the relationships between Sema4A expression and the essential molecules upregulated in psoriatic keratinocytes.

Page 21 Line 328 in the resubmitted manuscript:

“We were not able to reveal how Sema4A expression is regulated. Although we showed that downregulation of Sema4A is related to the abnormal cytokeratin expression observed in psoriasis, we could not determine the relationships between Sema4A expression and the essential molecules upregulated in psoriatic keratinocytes.”

(2) Using bone marrow chimeric mice, it has already been reported that hematopoietic cells contain keratinocyte stem cells. Therefore, their interpretation is not supported by the results of their bone marrow chimeric mice experiment, and it is essential to generate keratinocyte-specific Sema4A knockout mice and perform similar experiments to support their interpretation.

We value the reviewer’s insightful comment. We have assessed the expression of *Sema4a* in the epidermis of WT $\rightarrow$ KO chimeric mice using qRT-PCR. Our findings indicate that *Sema4a* expression levels in the epidermis of these mice are minimal (cycle threshold values of *Sema4a* ranged from 31.9 to not detected in WT $\rightarrow$ KO chimeric mice, whereas they ranged from 24.5 to 26.2 in WT $\rightarrow$ WT mice). Consequently, we believe that the impact of keratinocyte stem cells derived from WT-hematopoietic cells is limited in this model. We appreciate this opportunity to clarify our results and will consider the generation of keratinocyte-specific Sema4A knockout mice for future experiments to further substantiate our interpretation.

Page 11 Line 159 in the resubmitted manuscript:

“Since it has already been reported that bone marrow cells contain keratinocyte stem cells (Harris et al., 2004; Wu, Zhao, & Tredget, 2010), we confirmed that epidermis of mice deficient in non-hematopoietic Sema4A (WT $\rightarrow$ KO) showed no obvious detection of *Sema4a*, thereby ruling out the impact of donor-derived keratinocyte stem cells infiltrating the host epidermis (Figure 3-figure supplement 1A).”

Page 60: In the Figure legend of Figure 3-figure supplement 1A in the resubmitted manuscript:

“(A) *Sema4a* expression in the Epi of WT → WT mice and WT → KO mice ($n = 8$ for WT → WT, $n = 7$ for WT → KO).”

(3) Since Sema4A KO mice already have immunological and epidermal cell characteristics similar to psoriasis, albeit weak, it is possible that the nonspecific stimulus of simply topical IMQ may have appeared to exacerbate psoriasis. It is advisable to confirm whether a more psoriasis-specific stimulus, IL-23 administration, would produce similar results.

Thank you for your suggestion. Following your advice, we have analyzed IL-23-mediated psoriasis-like dermatitis. To induce the model, 20 μ l of phosphate-buffered saline containing 500 ng of recombinant mouse IL-23 was injected intradermally into both ears for 4 consecutive days. Unlike with the application of IMQ, there was no significant difference in ear thickness. However, H&E staining revealed that the epidermal thickness was significantly greater in KO mice compared to WT mice. Although a longer period of IL-23 induction might result in more pronounced ear swelling, we conducted this experiment over the same duration as the IMQ application experiment to maintain consistency. When we analyzed the T cells infiltrating the ears using flow cytometry, the proportion of IL-17A producing V γ 2 and DN γ δ T cells in CD3 fraction in the epidermis was significantly higher in Sema4A KO mice, consistent with the results from IMQ-induced psoriasis-like dermatitis.

The lack of significant difference in ear thickness changes with IL-23 administration might be due to IL-23 administration not reflecting upstream events of IL-23 production.

We consider that in psoriasis, the expression of Sema4A in keratinocytes is likely more important than in T cells. Therefore, it makes sense that the phenotype difference was more pronounced with IMQ, which likely has a greater effect on keratinocytes compared to IL-23.

Page 9 Line 137 in the resubmitted manuscript:

“Though the imiquimod model is well-established and valuable murine psoriatic model (van der Fits et al., 2009), the vehicle of imiquimod cream can activate skin inflammation that is independent of toll-like receptor 7, such as inflammasome activation, keratinocyte death and interleukin-1 production (Walter et al., 2013). This suggests that the imiquimod model involves complex pathway. Therefore, we subsequently induced IL-23-mediated psoriasis-like dermatitis (Figure2-figure supplement 2A), a much simpler murine psoriatic model, because IL-23 is thought to play a central role in psoriasis pathogenesis (Krueger et al., 2007; Lee et al., 2004). Although ear swelling on day 4 was comparable between WT mice and Sema4AKO mice (Figure2-figure supplement 2B), the epidermis, but not the dermis, was significantly thicker in Sema4AKO mice compared to WT mice (Figure2-figure supplement 2C). We found that the proportion of CD4 T cells among T cells was significantly higher in Sema4A KO mice compared to WT mice, while the proportion of V γ 2 and DN γ δ T cells among T cells was comparable between them (Figure 2-figure supplement 2D). On the other hand, focusing on IL-17A-producing cells, the proportion of IL-17A-producing V γ 2 and DN γ δ T cells in CD3 fraction in the epidermis was significantly higher in Sema4A KO mice, consistent with the results from imiquimod-induced psoriasis-like dermatitis. (Figure 2-figure supplement 2E).”

Page 24 Line 363 in the resubmitted manuscript: In the “Mice” section.

“To induce IL-23-mediated psoriasis-like dermatitis, 20 μ l of phosphate-buffered saline containing 500 ng of recombinant mouse IL-23 (BioLegend, San Diego, CA) was injected intradermally into both ears of anesthetized mice using a 29-gauge needle for 4 consecutive days.”

Page 58: In the Figure legend of Figure 2-figure supplement 2 in the resubmitted manuscript:

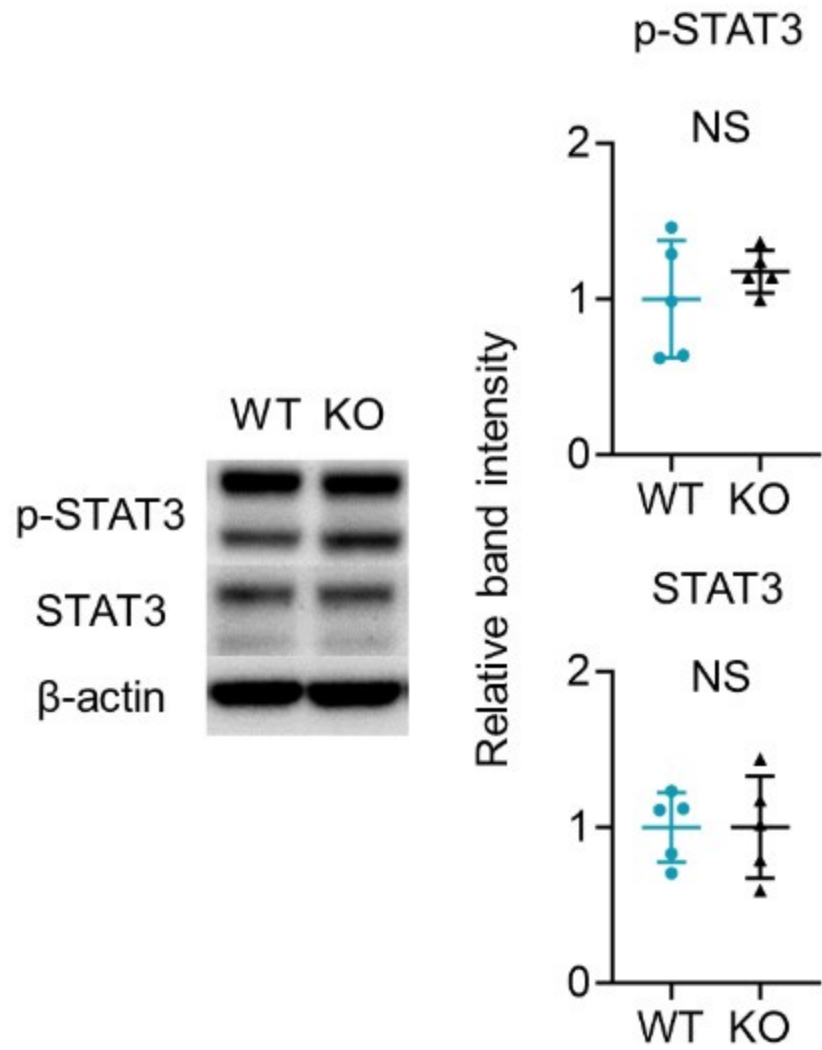
“IL-23-mediated psoriasis-like dermatitis is augmented in Sema4AKO mice.

(A) An experimental scheme involved intradermally injecting 20 μ l of phosphate-buffered saline containing 500 ng of recombinant mouse IL-23 into both ears of WT mice and KO mice for 4 consecutive days. Samples for following analysis were collected on Day 4. (B and C) Ear thickness (B) and Epi and Derm thickness (C) of WT mice and KO mice on Day 4 ($n = 12$ per group). (D and E) The percentages of $V\gamma 3$, $V\gamma 2$, $DN\gamma 8$, CD4, and CD8 T cells (D) and those with IL-17A production (E) in CD3 fraction in the Epi (top) and Derm (bottom) of WT and KO ears ($n = 5$ per group). Each dot represents the average of 4 ear specimens. B-E: * $p < 0.05$, ** $p < 0.01$. NS, not significant.”

(4) *How is STAT3 expression in the epidermis crucial in the pathogenesis of psoriasis in Sema4AKO mice?*

We appreciate your insightful comment. In our study, given the established role of activated STAT3 in psoriasis, we investigated both total STAT3 and phosphorylated STAT3 (p-STAT3) levels in the naive epidermis of WT and Sema4AKO mice (See the figure below). Our findings indicate that STAT3 activation does not occur in the epidermis of Sema4AKO mice. Therefore, we speculated that the hyperkeratosis observed in Sema4AKO mice is due to aberrant mTOR signaling rather than STAT3 activation. STAT3 may be relevant to other pathways independent of Sema4A signaling, or it may function as a complex with other molecules in the Sema4A signaling.

Author response image 4.



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